

Command Reference

NP-F309*

3 inch Face Mount Printer

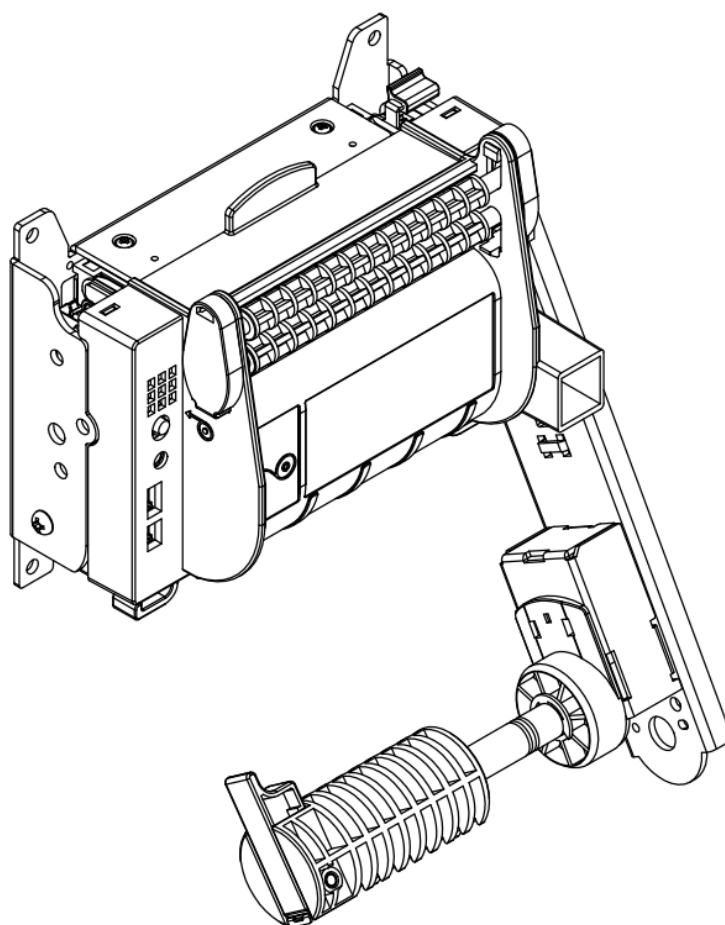


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1. Application

This document describes commands and character code tables of “NP-F309*” (3 inch compact KIOSK face mount printer).

Please refer to Product Specifications of “NP-F309*” for specifications, precautions and the revision history.

[Notes]

- This document is a command reference.
- All specifications described in this document are subject to change for software change or improvement of this product without prior notice.

2. Command

2.1 Command table

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Commands with “◆” are commands whose setting will be initialized (being returned to initial settings) by the command “[Printer initialization] 《ESC @》”. Please refer to details of each command regarding initial settings.

[Notes]

- “[F/W]” indicates firmware hereinafter.

2.2 Command details

1) [Horizontal tab] 《HT》

Code : [09] h

Moving the printing position to the next horizontal tab position.

- Horizontal tab is set by the command “[Horizontal tab position setting] 《ESC D n1 n2 --- NULL》”.
- “Horizontal tab position” by the default setting is set with the intervals by 8 characters of Font A. They will be the position of 9th digit, 17th digit, 25th digit, 33rd digit and 41st digit.
- When the next “Horizontal tab” is not set, this command will be disregarded.

2) [Print and line feeding] 《LF》

Code : [0A] h

Printing out data in the print line buffer and feeding lines based on preset line feeding amount.

3) [Carriage return] 《CR》

Code : [0D] h

- This command will be disregarded.

4) [Software reset] 《DC1》

Code : [11] h

Restarting the firmware by the same process as the process during the power turning ON.

- Since this command is stored in the internal receiving input buffer and will be executed sequentially, timing of receiving the command and execution differs.
- While the cutter is operating, software reset will be activated after the cutter operation finishes.

5) [Ruled line printing permission] 《DC3 +》 ◆

Code : [13] h + [2B] h

Permitting the printer to print the data inside the ruled line buffer. —

- Ruled line printing is prohibited when the power turning ON or the printer initialization is completed.
- Ruled lines will not overlap with “Page layout data”, “Barcode”, “QR code”, “Fixed bit image” or “Raster bit image”.

6) [Ruled line printing prohibition] 《DC3 -》

Code : [13] h + [2D] h

Prohibiting the printer from printing the data inside the ruled line buffer.

- The data in the ruled line buffer will never be printed after prohibition.
- Ruled line printing will become prohibited by turning the power ON by the printer initialization.

7) [Ruled line buffer A selection] 《DC3 A》

Code : [13] h + [41] h

Selecting the ruled line buffer A.

- This printer has 2 individual ruled line buffers, buffer A and buffer B.
- Buffer A has been selected by the default setting.

14) [Ruled line image data writing] 《DC3 V Dn》

Code : [13] h + [56] h + Dn

Writing the 1 dot line image data into the specified ruled line buffer.

- “Dn” is raster bit image data.
- Regarding raster bit image data (Dn), “1” is filled for the bit to be printed and “0” for the bit not to be printed.
- Number of “Dn” data will be 72 bytes.

15) [Changing the last part of the barcode printing command] 《ESC RS c n》 ◆

Code : [1B] h + [1E] h + [63] h + n ※ [n=00, 80] h

Changing the last part of the command “[Barcode print] 《GS k n Dn NUL》” by “n”.

- “n” has following meanings.

n (hex)	The last part
00	[00] h
80	[FF] h

- Default value of “n” is “ [00] h”.

16) [Character right space amount setting] 《ESC SP n》 ◆

Code : [1B] h + [20] h + n ※ [00≤n≤20] h

Setting the right space amount of the character by dots (1/203 inch unit).

- The amount of right space will become double when selecting double width mode.
- Default value of “n” is “ [00] h”.

17) [Specifying print modes all at once] 《ESC ! n》 ◆

Code : [1B] h + [21] h + n ※ [00≤n≤FF] h

Specifying print modes all at once.

- Each bit of “n” has following meanings.

bit	Function	Value	
		0	1
bit0	Character font	Font A	Font B
bit1	Undefined	-	-
bit2	Undefined	-	-
bit3	Enhanced	Releasing	Specifying
bit4	Double height	Releasing	Specifying
bit5	Double width	Releasing	Specifying
bit6	Undefined	-	-
bit7	Underline	Releasing	Specifying

- If double height and double width are set at the same time, character with double-height and double-width will be formed.
- All printed characters will be underlined except for 90° clock-wise rotated characters and spaces created by the command “[Horizontal tab] 《HT》”.
- Underline height is determined by the value set in the command “[Underline specifying / releasing] 《ESC - n》”. Default value is “1 dot” height.
- Only enhanced print control is effective during Kanji mode.
- Mixed different character sizes (e.g. double width and normal size, etc.) can be printed.
- Default value of “n” is “ [00] h”.

18) [Ruled line overlap mode] 《ESC # n》 ◆

Code : [1B] h + [23] h + n

Selecting the mode for overlapping ruled lines with characters.

- "n" has following meanings.

n (hex)	Mode
00	OR overlap mode
01	XOR overlap mode

- Default value of "n" is " [00] h".
- Ruled lines will not overlap "Barcode", "QR code", "Fixed bit image" or "Raster bit image".

19) [Absolute position setting] 《ESC \$ n1 n2》 ◆

Code : [1B] h + [24] h + n1 + n2

※ [00 ≤ n1 ≤ FF] h

※ [00 ≤ n2 ≤ 02] h

Specifying the print start position from the head of the line by the number of dots (by 1/203 -inch position).

- The quotient obtained by dividing the value of numbers of dots for the printing starting position by 256 shall be "n2" and the remainder by it shall be "n1".
- Printing starting position is (n1 + n2 × 256) from the head of the line.
- Disregarding the setting exceeding the end of the line.
- When this command is received in the middle of the line, it will be effective even when it does not pass the current position.

20) [Download characters set specifying / releasing] 《ESC % n》 ◆

Code : [1B] h + [25] h + n

※ [00 ≤ n ≤ FF] h

Specifying / releasing the characters set to be downloaded.

- Only the least significant bit (bit0) is effective to "n". "bit0" has following meanings.

bit0	Function
0	Releasing download characters set
1	Specifying download characters set

- Default value of "n" is " [00] h".

21) [Download characters definition] 《ESC & s n m a Dn》

Code : [1B] h + [26] h + s + n + m + a + Dn

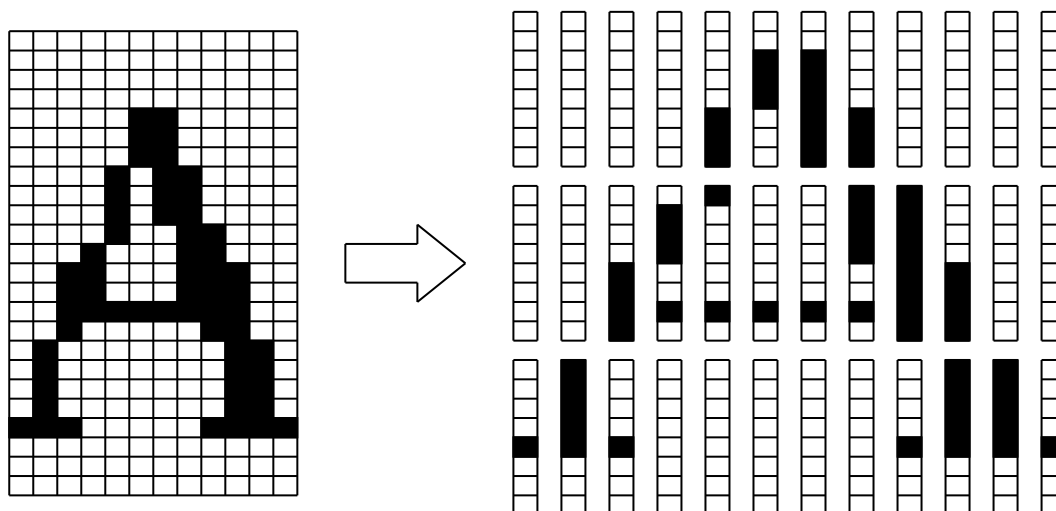
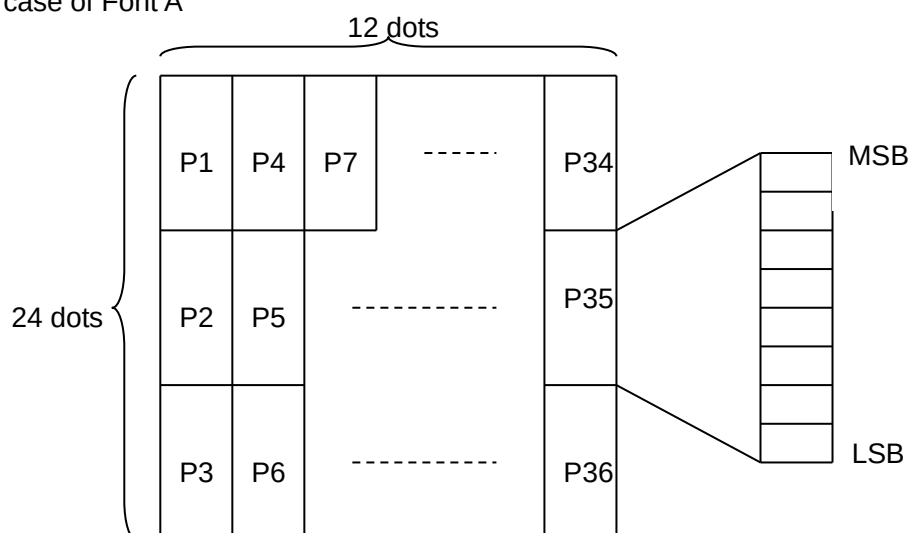
- ※ [s=03] h
- ※ [20≤n≤7E] h
- ※ [20≤m≤7E] h
- ※ Font A [01≤a≤0C] h
- ※ Font B [01≤a≤09] h

Defining download characters of alphanumeric characters.

- “s” indicates the number of bytes to be defined in vertical direction while “a” indicates the number of dots in horizontal direction.
- “n” indicates the starting character code and “m” indicates the finishing character code. (When defining only one character definition, “n” = “m”.)
- Definable characters are 95 characters from " [20] h to [7E] h" in ASCII code.
- “Dn” is the data to be defined and indicates the pattern on “a” dots in the horizontal direction from the left edge. Remaining pattern on the right side will be filled with spaces.
- Download characters once defined by this command remain effective until executing the command “[Software reset] 《DC1》”, executing reset by pressing the RESET button or turning OFF the power.
- Redefinition only redefines the designated area.

<Reference>

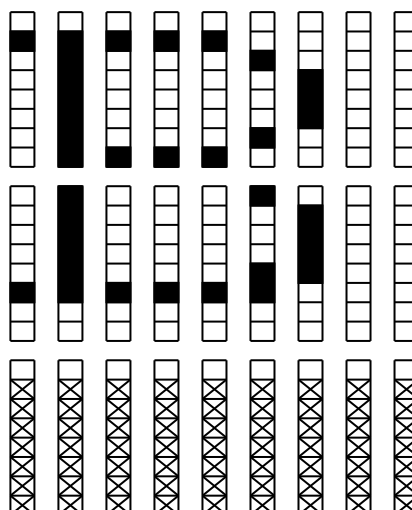
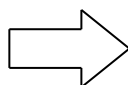
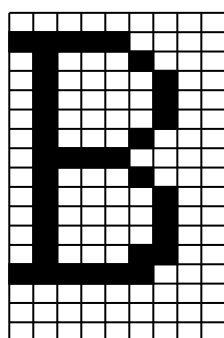
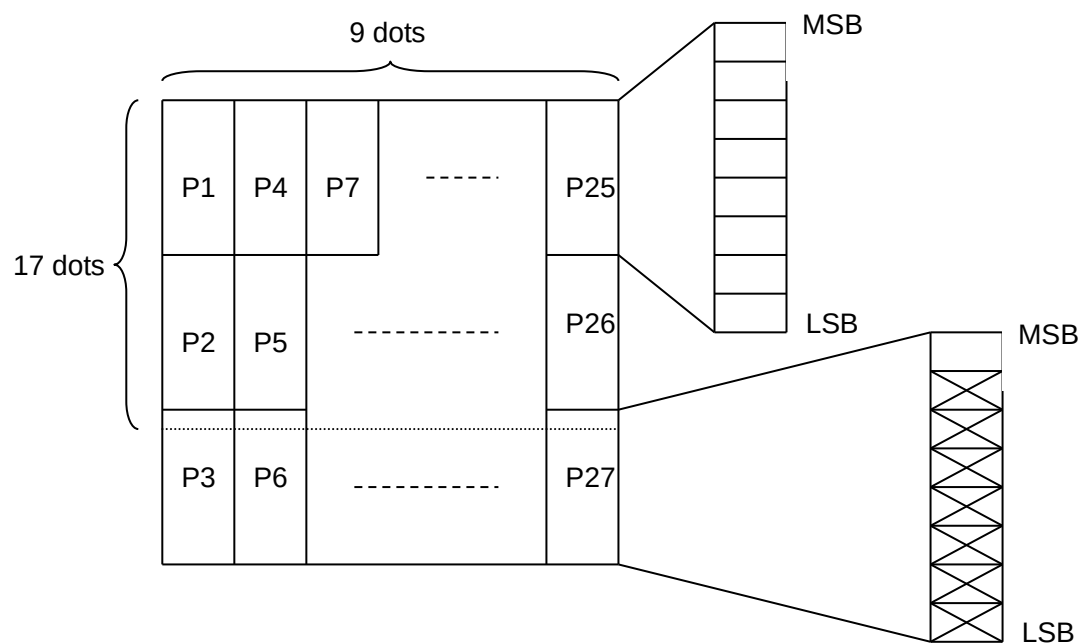
In case of Font A



```
P1= [00] h,P4= [00] h,P7= [00] h,P10= [00] h,...
P2= [00] h,P5= [00] h,P8= [0F] h,P11= [72] h,...
P3= [08] h,P6= [F8] h,P9= [08] h,P12= [00] h,...
```

<Reference>

In case of Font B



P1= [40] h,P4= [7F] h,P7= [41] h,P10= [41] h, ...
P2= [04] h,P5= [FC] h,P8= [04] h,P11= [04] h, ...
P3= [00] h,P6= [00] h,P9= [00] h,P12= [00] h, ...

22) [Bit image mode specifying] 《ESC * m n1 n2 Dn》

Code : [1B] h + [2A] h + m + n₁ + n₂ + Dn ※ [m=(described in the table below)] h
 ※ [00 ≤ n₁ ≤ FF] h
 ※ [00 ≤ n₂ ≤ 02] h

Printing the data in bit image with resolution (bit image mode) specified by “m”.

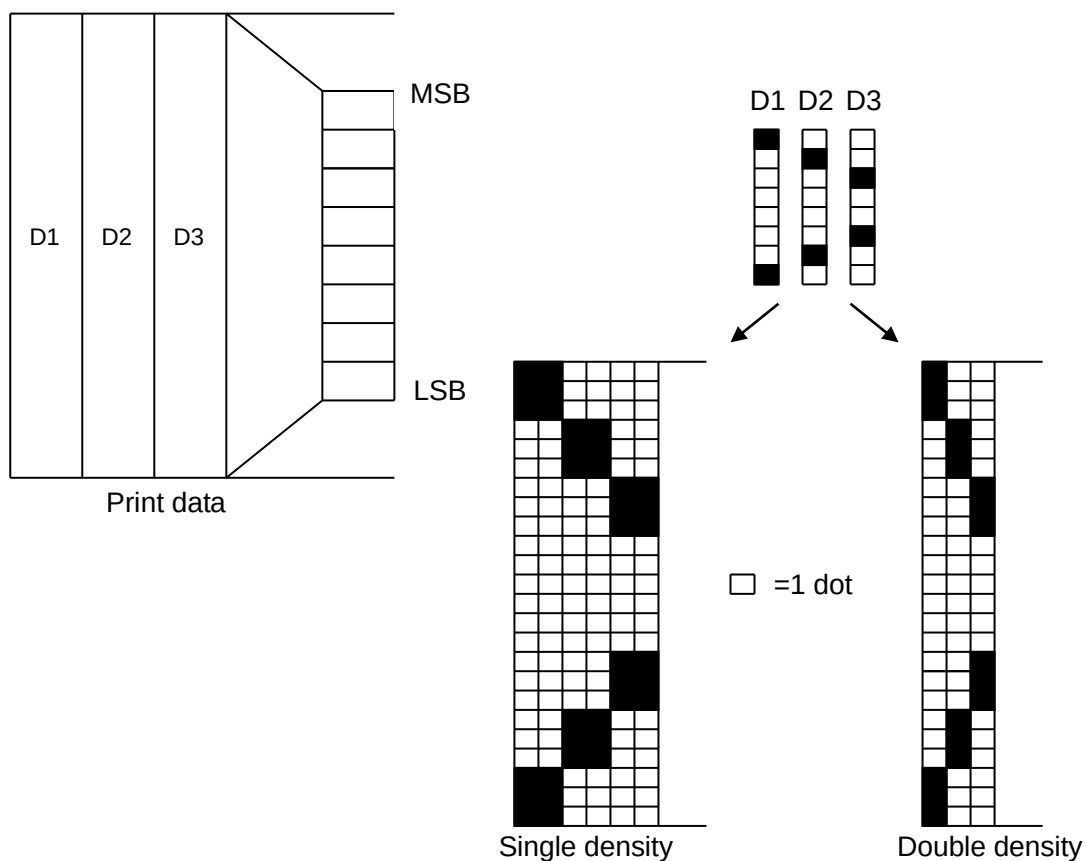
- The quotient obtained by dividing the value of numbers of dots by 256 shall be "n2" and the remainder by it shall be "n1".
- Total dots to be printed as a bit image mode are " $n1 + (256 \times n2)$ ".
- If bit image data (Dn) which exceeds the print area is input, exceeding data will be disregarded.
- Bit image data (Dn) is "1" for bits to be printed and "0" for bits not to be printed.
- Bit image modes are described in the table below.

<Standard mode>

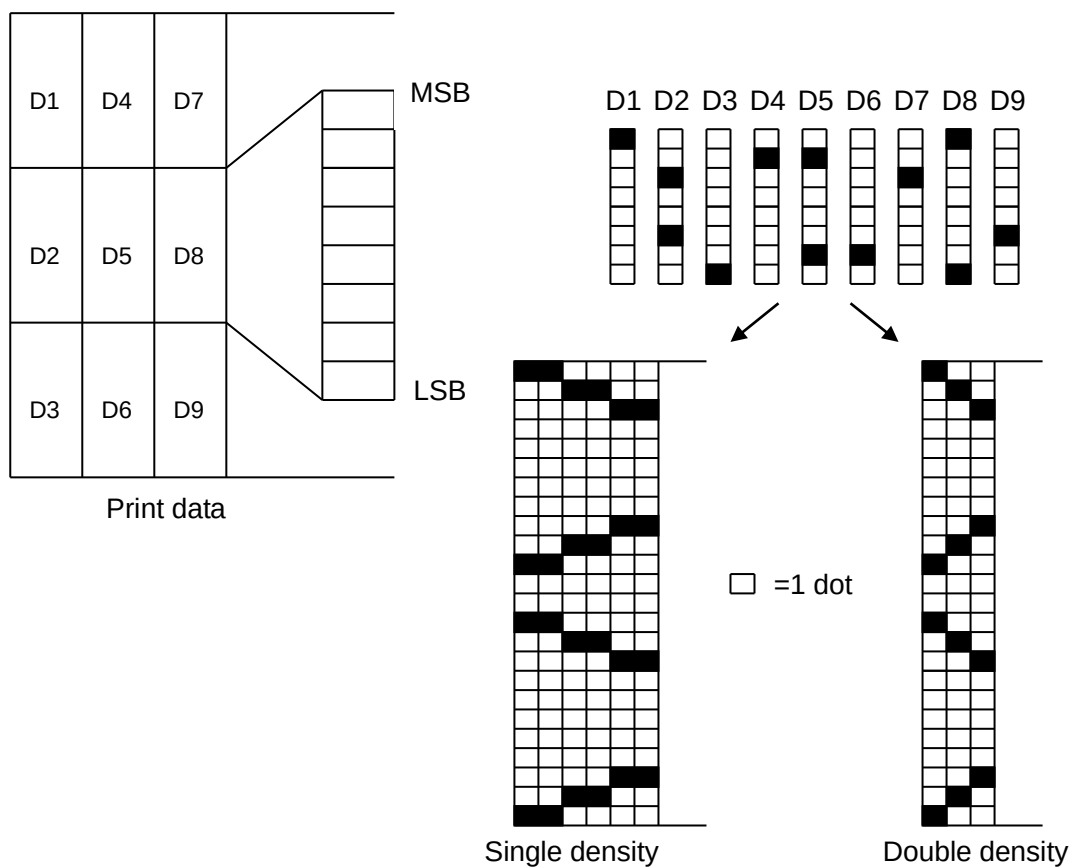
m (hex)	Bit image mode	Vertical direction		Horizontal direction	
		Number of dots	Dots density	Dots density	Maximum number of dots
00	8 dots single density	8	67DPI	101DPI	288
01	8 dots double density	8	67DPI	203DPI	576
20	24 dots single density	24	203DPI	101DPI	288
21, 23	24 dots double density	24	203DPI	203DPI	576

<Relationship between the bit image data and printed dots>

• 8 dots bit image



• 24 dots bit image



23) [Underline specifying / releasing] 《ESC - n》 ◆

Code : [1B] h + [2D] h + n

※ [00 ≤ n ≤ 02] h

Specifying / Releasing underline.

- Underline is attached to the entire printing character width except for the area skipped by the command "[Horizontal tab] 《HT》". Furthermore, underline is not attached to 90° clockwise rotated characters.
- This command is not effective during Kanji mode.
- Types of underlines designated by the value of "n" are described below.

n (hex)	Type
00	Releasing underline
01	Specifying one dot height underline
02	Specifying two dots height underline

- Default value of "n" is "[00] h".

24) [1/6 inch line feeding specifying] 《ESC 2》

Code : [1B] h + [32] h

Specifying 1/6 inch line feeding.

25) [Line feeding amount setting to the smallest paper feeding pitch unit] 《ESC 3 n》 ◆

Code : [1B] h + [33] h + n

※ [00 ≤ n ≤ FF] h

Setting the line feeding amount for a line to n/203-inch.

- Even when the smaller value than character's height is set, it will execute line feeding for the amount of character's height.
- Default value of "n" is "[22] h" (34 dots).

26) [Printer initialization] 《ESC @》

Code : [1B] h + [40] h

Clearing data stored in the print line buffer and initializing each setting to the initial setting (to the default status).

- Not clearing the data stored in the internal receiving input buffer.
- Not clearing the data stored in the ruled line buffer.
- It is stored in the receiving buffer and will be executed sequentially.

27) [Back feeding] 《ESC B n》

Code : [1B] h + [42] h + n

※ [00 ≤ n ≤ FF] h

Feeding paper to the reverse direction.

- Setting the feeding amount by "n" dot(s) line. In case of setting "[00] h", it is never fed.
- Please DO NOT enter this command in a row for preventing paper jams.
When feeding reversely more than "[FF] h" dot(s), please use the command "[Back feeding (mm unit)] 《GS J n》".
- Some deviation may occur due to taking the backlash.
- If there is print data in the print line buffer, back feeding will be performed after printing.
- Please DO NOT make the tip of paper exceed the back feeding limitation.
(Please refer to "2.4 Cutter specifications" in Product Specifications.)

28) [Horizontal tab position setting] 《ESC D n1 n2 --- NUL》 ◆

Code : [1B] h + [44] h + n1 + n2 + --- + [00] h ※ [00 ≤ n ≤ FF] h

Setting the horizontal tab position.

- "n" indicates digits from the head to the tab position.
In this case, it shall be "[n = tab set position - 1]".
- The tab position is set at the point of "character width × n" from the head of the line. Character width in this case includes right space amount of the character and it will be doubled when the double width enlargement is set.
- Maximum 32 tab positions can be set and positions set exceedingly will be disregarded.
- "《ESC D NUL》" clears all set tab positions. The command "[Horizontal tab] 《HT》" after clearing will be disregarded.
- The default setting is set with the intervals by 8 characters of Font A. They will be the position of 9th digit, 17th digit, 25th digit, 33rd digit and 41st digit.

29) [Enhanced print specifying / releasing] 《ESC E n》 ◆

Code : [1B] h + [45] h + n ※ [00 ≤ n ≤ FF] h

Specifying / Releasing enhanced print.

- Only LSB (least significant bit "bit0") is available for "n".
- "bit0" has following meanings.

bit0	Descriptions
0	Releasing enhanced print
1	Specifying enhanced print

- When executing enhanced print, print result may be deformed.
- Default value of "n" is "[00] h".

30) [Double strike print specifying / releasing] 《ESC G n》 ◆

Code : [1B] h + [47] h + n ※ [00 ≤ n ≤ FF] h

Specifying / Releasing double strike print.

- Only LSB (least significant bit "bit0") is available for "n".
- "bit0" has following meanings.

bit0	Descriptions
0	Releasing double strike print
1	Specifying double strike print

- Since this printer cannot generate "double strike print" expression, printing results will be the same as enhanced print.
- Default value of "n" is "[00] h".

31) [Print and paper feeding by the smallest pitch unit] 《ESC J n》

Code : [1B] h + [4A] h + n ※ [00 ≤ n ≤ FF] h

Printing the data inside the print line buffer and feeding paper by n/203 inch.

- Line feeding amount does not remain.
- Beginning of the line shall be a next print starting position.
- Executing line feeding for the amount of character's height, even if the smaller value of character's height is set.

32) [International characters selection] 《ESC R n》 ◆

- * Effective only when selecting either "Domestic code table" or "Overseas code table" by the command "[Character code table selection] 《ESC t n》".

Code : [1B] h + [52] h + n ※ [00 ≤ n ≤ 0A] h

Selecting international characters.

- "n" has following meanings.

n (hex)	Character setting
00	U.S.A.
01	France
02	Germany
03	U.K.
04	Denmark I
05	Sweden
06	Italy
07	Spain
08	Japan
09	Norway
0A	Denmark II

- Default value of "n" is "[08] h".

33) [90° clockwise rotated characters specifying / releasing] 《ESC V n》 ◆

Code : [1B] h + [56] h + n ※ [00 ≤ n ≤ 01] h

Specifying / Releasing 90° clockwise rotated characters.

- "Underline specifying" setting will become ineffective for 90° clockwise rotated characters
- "n" has following meanings.

n (hex)	Descriptions
00	Releasing 90° clockwise rotated characters
01	Specifying 90° clockwise rotated characters

- Default value of "n" is "[00] h".

34) [Relative position setting] 《ESC ¥ n1 n2》 ◆

Code : [1B] h + [5C] h + n1 + n2 ※ [00 ≤ n1 ≤ FF] h
 ※ [00 ≤ n2 ≤ FF] h

Setting the print start position by the number of dots (1/203-inch unit) counted from the current position.

- The quotient obtained by dividing the value of numbers of dots for the printing starting position by 256 shall be "n2" and the remainder by it shall be "n1".
- It will be moved from the current position to the right by specifying it with plus and to the left by specifying it with minus.
- When "N" dots is specified to the plus direction, it will be "(n1" + "n2" × 256)".
- When "N" dot is specified to the minus direction, it will be set by using the complementary number of "N".

$$N \text{ dots} = 65536 - N$$

- Setting exceeding the end of the line will be disregarded.

35) [Position alignment] 《ESC a n》 ◆

Code : [1B] h + [61] h + n ※ [00 ≤ n ≤ 02]

Aligning the print data in a line at the specified position.

- "n" has following meanings.

n (hex)	Position
00	Left alignment
01	Centering
02	Right alignment

- Effective only when it is input at the beginning of the line.
- Default value of "n" is "[00] h".

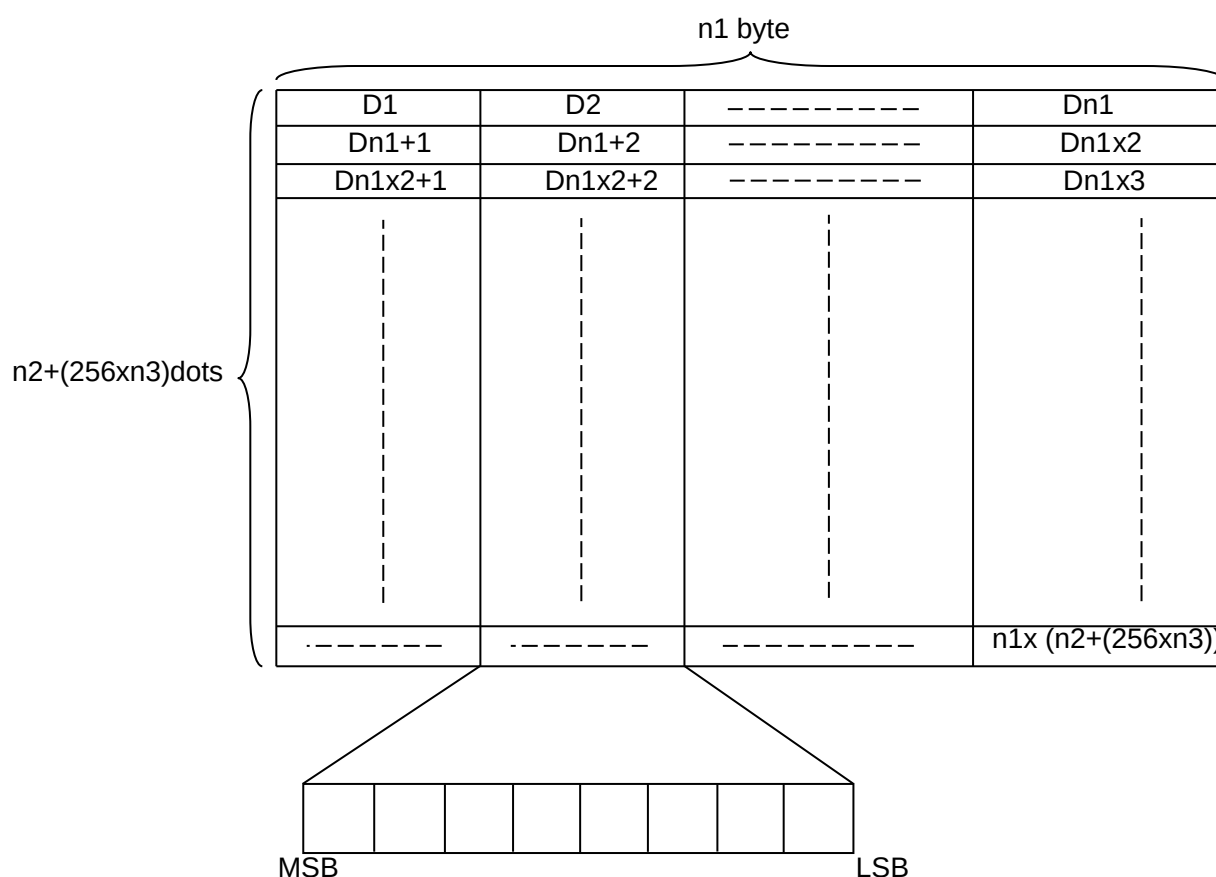
36) [Raster bit image] 《ESC b n1 n2 n3 Dn》

Code : [1B] h + [62] h + n1 + n2 + n3 + Dn

- ※ [01 ≤ n1 ≤ 48] h
- ※ [00 ≤ n2 ≤ FF] h
- ※ [00 ≤ n3 ≤ FF] h

Printing data in raster bit image.

- “Dn” is raster bit image data.
- Printing raster bit image whose size is “n1” bytes wide and “n2 + (256 x n3)” dot lines height.
- The total number of bytes required for raster bit image data (Dn) is {n1 x (n2 + (256 x n3))}.
- Raster bit image data (Dn) exceeding the print area will be disregarded.
- Regarding raster bit image data (Dn), “1” is for bits to be printed and “0” is for bits not to be printed.
- If a print stop error (Please refer to “4.2 Processing errors” in the Product Specifications.) occurs, all of following data after the print stop will be disregarded. Please transmit all data from the beginning when printing is restarted again.
- Relation between raster bit image data and printed dots are as follows.



- Printing by this command will be executed after print data storing regardless of the setting made by using the command “[Print start / finish setting] 《GS G n》”.

37) [FEED button effective / ineffective] 《ESC c 5 n》 ◆

Code : [1B] h + [63] h + [35] h + n ※ [00 ≤ n ≤ FF] h

Enabling or disabling the FEED button.

- Only LSB (least significant bit “bit0”) is available for “n”.
- “bit0” has following meanings.

bit0	Descriptions
0	Enabling the FEED button
1	Disabling the FEED button

- Default value of “n” is “ [00] h”.

38) [Print and “n” lines feeding] 《ESC d n》

Code : [1B] h + [64] h + n ※ [00 ≤ n ≤ FF] h

Printing the data inside the print line buffer and feeding paper for “n” lines.

- Specified amount will not remain.
- Beginning of the line shall be the next printing start position.
- If there are printing data, it definitely performs line feeding for the character's height.
- When “n lines × line feeding amount” becomes more than 100mm, line feeding amount will become 100mm.

39) [Full cut] 《ESC i》

Code : [1B] h + [69] h

Executing full cut of paper.

- Only effective at the beginning of a line.

40) [Partial cut] 《ESC m》

Code : [1B] h + [6D] h

Executing partial cut of paper. (The center part of paper will remain uncut.)

- Only effective at the beginning of a line.

41) [QR code printing (Model2)] 《ESC q S E V M》

Code : [1B] h + [71] h + S + E + V + M + n1 + n2 + Dn

Descriptions regarding parameters

(1) S : Module size

- Specifying the size of one QR code module by the number of dots of the printer.
- Selectable size is between 1 and 20 dots.
- When invalid size is specified, it will become 4 dots. (Readability is not guaranteed for module sizes 1, 2 and 3 because of the resolution of this printer mechanism.)

(2) E : Error correction level

- Selecting the error correction level for recovering symbols.
- Selectable values are as below.
- When invalid value is selected, error correction level will be "L".

E	Error correction level	Recovering ability (%)
0	L	7
1	M	15
2	Q	25
3	H	30

(3) V : Model number (version)

- Selecting the version number from 0 to 40 for deciding the number of QR code modules.
- Version 1 has the smallest number of modules and version 40 has the largest.
- When selecting "0" for the version, the smallest version for the data size will be selected automatically.
- Version 1 has 21 × 21 modules and version 40 has 177 × 177 modules. (4 for both the vertical direction and the horizontal direction will be increased by 1 version number increase.)
- In case an ineffective value is specified, it will be "0".
- When the data string with the larger size than that of specified version, it will be changed for expanding the number of modules before printing.

(4) M : Masking pattern

- Selectable masking pattern is between 0 and 8.
- Optimal masking pattern will be set for masking pattern 0. ^{*1}
- When an invalid value is selected, masking pattern will be 5.

^{*1} : Although the masking pattern optimization process is made according to the QR code specifications, it will take time. Therefore, please use the model number 10 or less when using optimization.

Please refer to "[Processing time for masking pattern optimization]" on the next page for the processing time.

(5) n1 n2 : The number of data bytes

- Please specify the number of data bytes by (n2 × 256 + n1).

(6) Dn : Character string data

- Please select the character code to print.
- Please input Japanese Kanji data by Shift JIS code.

Restrictions

When exceeding the print area, the QR code will not be printed.

[Processing time for masking pattern optimization]

1. Printing condition

Size (S)	Correction level (E)	Masking pattern (M)	Number of bytes (B)	Data (Dn)
4	L	0	20	漢漢漢漢漢漢漢漢漢漢

2. Processing time

Model No. (V)	1	2	3	4	5	6	7	8	9	10
Time (sec.)	0.4	0.5	0.7	0.9	1.0	1.3	1.5	1.7	2.0	2.3

Model No. (V)	11	12	13	14	15	16	17	18	19	20
Time (sec.)	2.7	3.0	3.4	3.8	4.2	4.7	5.1	5.7	6.2	6.7

Model No. (V)	21	22	23	24	25	26	27	28	29	30
Time (sec.)	7.3	7.8	8.5	9.1	9.7	10.5	11.2	11.8	12.6	13.4

Model No. (V)	31	32	33	34	35	36	37	38	39	40
Time (sec.)	14.1	15.0	15.8	16.7	17.5	18.4	19.4	20.4	21.3	22.3

[Notes]

- Processing time depends on the model number.
- Please note that the processing time when it is optimized is approx. 10 times long as when the masking pattern is fixed (when not optimized).
- Times described above are just for reference.

42) [Printer information transmission] 《ESC s n》

Code : [1B] h + [73] h + n

Transmitting printer information.

- “n” and details regarding the returned data are as follows.

n (hex)	Printer information type	Returned data format	Returned data length
02	Model information	Variable-length characters string (terminal NULL = “ [00] h”)	Max. 32 bytes
03	F/W version information	Fixed-length characters string	8 bytes
04	Boot version information	Fixed-length characters string	8 bytes
05	Switch setting information	Fixed-length hexadecimal data (MS1 setting + MS2 setting + “ [00] h” + “ [00] h”)	4 bytes
1C	Checksum information	Fixed-length hexadecimal data	2 bytes

Format of the returned data

[FF] h + n (*1) + returned data

*1 : It is “n” specified by the command.

- Please refer to “[4.1.1 Switch setting]” in the Product Specifications when using XON / XOFF control under the interface set to serial.
- Please refer to “[3.2 6) Transmitted data]” in the Product Specifications for the transmission format.

43) [Character code table selection] 《ESC t n》 ◆

Code : [1B] h + [74] h + n

※ [00 ≤ n ≤ 07] h

Selecting the character code table.

- “n” has following meanings.

n (hex)	Font table
00	Overseas character code
01	Domestic character code
02	Code Page 858
03	Code Page 1250
04	Code Page 1251
05	Code Page 1252
06	Code Page 1254
07	User Code Page

- Default value of “n” is “ [01] h”.

44) [Printer status transmission] 《ESC v》

Code : [1B] h + [76] h

Transmitting the latest printer status.

- Transmitted status is one byte data. Please refer to “[4.2 Processing errors 1) Details of error detection]” in the Product Specifications. Furthermore, please refer to “[3.2 6) Transmitted data]” in the Product Specifications for the transmission format.
- One byte data will be transmitted only after confirming that the host is ready for receiving the data (CTS signal : SPACE status).
- When the host is not ready for receiving the data (CTS signal : MARK status), the printer will wait until it turns to be ready.
- This command is effective only with serial interface.
- Please be sure to issue this command before transmitting the print data.
(It will be stored in the internal receiving input buffer and will be executed sequentially.)
- It is ready for receiving the data except when internal receiving input buffer is full.
- Please refer to “[4.1.1 Switch setting]” in the Product Specifications when using XON / XOFF control under the interface set to serial.

45) [Inverted character specifying / releasing] 《ESC { n》 ◆

Code : [1B] h + [7B] h + n ※ [00≤n≤FF] h

Specifying / Releasing inverted print.

- Only LSB (least significant bit “b0”) is available for “n”.
- “b0” has following meanings.

bit0	Descriptions
0	Releasing inverted character
1	Specifying inverted character

- This command is only effective when it is assigned at the beginning of a line.
- Default value of “n” is “ [00] h”.

46) [Partition drive selection] 《GS % n》

Code : [1D] h + [25] h + n ※ [00≤n≤04] h

Selecting the partition drive type.

- “n” has following meanings.

n (hex)	Partition	Number of dots being energized at one time
00	Optimization (default setting)	144
01	Fixed without partition	576
02	Fixed with dual partitions	288
03	Fixed with four partitions	144
04	Optimization (default)	144

- Default value of “n” is “ [04] h”.
- If it is out of range, this command will be disregarded and it will not be changed.
- Results set by “ [00] h” and “ [04] h” are same.

47) [User code page registration] 《GS & n Dn》

Code : [1D] h + [26] h + n + Dn ※ [00≤n≤01] h

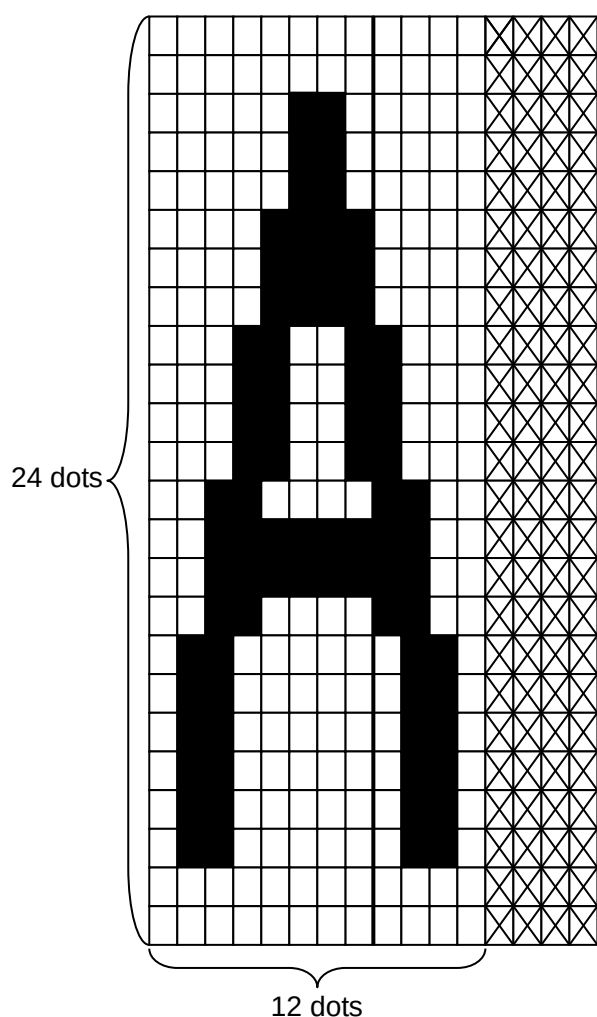
Registering user code page.

n (hex)	Types of the font to be registered
00	User Code Page Font A (12 x 24)
01	User Code Page Font B (9 x 17)

- “Dn” indicates the font data to be registered. Data format is raster format.
Please select “ [00] h” for all data not being registered because the data for 224 characters (10,752 bytes) are always required.
[horizontally 2 bytes x vertically 24 bytes] x 224 characters (“ [20] h” to “ [FF] h”)
- They can be registered to “ [7F] h” and “ [FF] h”.

<Reference>

In case of Font A

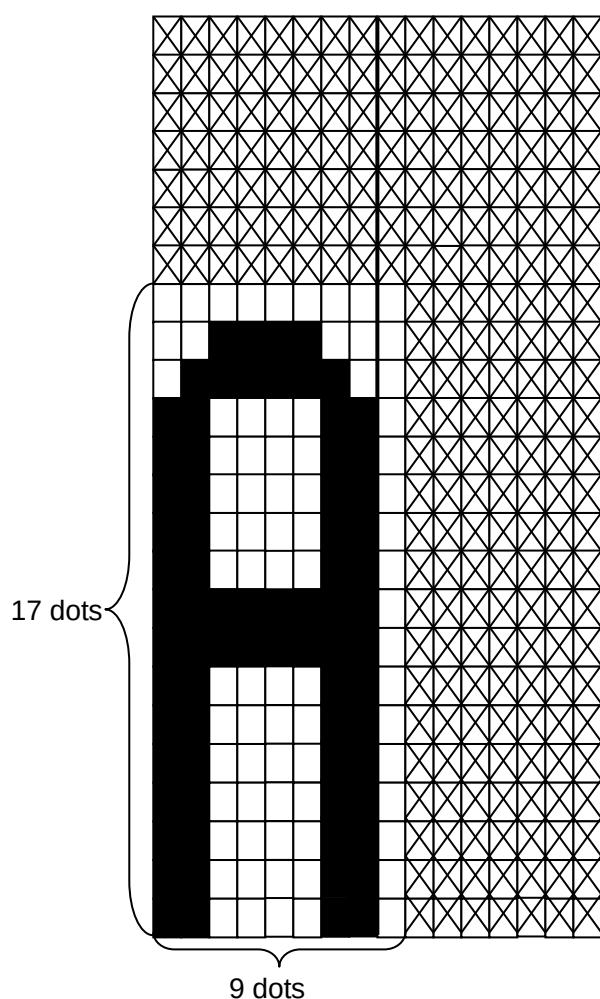


d1 = [00] h,	d2 = [00] h
d3 = [00] h,	d4 = [00] h
d5 = [06] h,	d6 = [00] h
d7 = [06] h,	d8 = [00] h
d9 = [06] h,	d10 = [00] h
d11 = [0F] h,	d12 = [00] h
d13 = [0F] h,	d14 = [00] h
d15 = [0F] h,	d16 = [00] h
d17 = [19] h,	d18 = [80] h
d19 = [19] h,	d20 = [80] h
d21 = [19] h,	d22 = [80] h
d23 = [19] h,	d24 = [80] h
d25 = [30] h,	d26 = [C0] h
d27 = [3F] h,	d28 = [C0] h
d29 = [3F] h,	d30 = [C0] h
d31 = [30] h,	d32 = [C0] h
d33 = [60] h,	d34 = [60] h
d35 = [60] h,	d36 = [60] h
d37 = [60] h,	d38 = [60] h
d39 = [60] h,	d40 = [60] h
d41 = [60] h,	d42 = [60] h
d43 = [60] h,	d44 = [60] h
d45 = [00] h,	d46 = [00] h
d47 = [00] h,	d48 = [00] h

☒ : Since these are dummy data cells, please fix them by 0 when registering it.

<Reference>

In case of Font B



d1 = [00] h, d2 =00] h
d3 = [00] h, d4 =00] h
d5 = [00] h, d6 =00] h
d7 = [00] h, d8 =00] h
d9 = [00] h, d10=00] h
d11= [00] h, d12=00] h
d13= [00] h, d14=00] h
d15= [00] h, d16=00] h
d17= [3C] h, d18=00] h
d19= [7E] h, d20=00] h
d21= [C3] h, d22=00] h
d23= [C3] h, d24=00] h
d25= [C3] h, d26=00] h
d27= [C3] h, d28=00] h
d29= [C3] h, d30=00] h
d31= [FF] h, d32=00] h
d33= [FF] h, d34=00] h
d35= [C3] h, d36=00] h
d37= [C3] h, d38=00] h
d39= [C3] h, d40=00] h
d41= [C3] h, d42=00] h
d43= [C3] h, d44=00] h
d45= [C3] h, d46=00] h
d47= [C3] h, d48=00] h

☒ : Since these are dummy data cells, please fix them by 0 when registering it.

48) [Black and white reverse print specifying / releasing] 《GS B n》

Code : [1D] h + [42] h + n ※ [00≤n≤FF] h

Specifying / Releasing black and white reverse print.

- Only LSB (least significant bit “bit0”) is available for “n”.
- “bit0” has following meanings.

bit0	Function
0	Releasing black and white reverse print
1	Specifying black and white reverse print

- Black and white reverse print is applicable to built-in characters and downloaded characters.
- Black and white reverse print is also applied to the right side space of the set character specified by the command “[Character right space amount setting] 《ESC SP n》”. It does not affect the skipped space created by bit image, raster bit image, fixed bit image, barcode, HRI characters, the command “[Horizontal tab] 《HT》”, the command “[Absolute position setting] 《ESC \$ n1 n2》” or the command “[Relative position setting] 《ESC ¥ n1 n2》”.
- It does not affect spaces between lines.
- Since setting the black and white reverse print has higher priority than underline setting, underline will not be added to black and white reverse characters even when underline is set. However, underline setting remains effective.
- When black and white reverse print is performed, printing result may be deformed.
- Default value of “n” is “ [00] h”.

49) [Specifying partition drive optimization condition] 《GS C n》

Code : [1D] h + [43] h + n ※ [00≤n≤03] h

Specifying how partition drive optimization is judged.

- “n” has following meanings.

n (hex)	Condition type	Printing speed	Peak current
00	High speed	High ↓ Low	High ↓ Low
01	Standard		
02	Power saving (low)		
03	Power saving (high)		

- Default value of “n” is “ [01] h”.
- If it is out of range, this command will be disregarded and will not change.

50) [Print start / finish setting] 《GS G n》

Whether the print operation has been finished or not can be judged by monitoring the 7th bit of the printer status.

1. Print start / finish setting (n = [00] h, [02] h, [10] h, [12] h)
 - If a print stop error occurs during printing the data sandwiched between the print start setting and the print finish setting, the print data up to the print finish setting is not disregarded.
 - (1) [Print start setting (without JOB ID)]
 - Code : [1D] h + [47] h + [02] h
 - Setting "1" to the "bit7" of the printer status.
 - (2) [Print finish setting (without JOB ID)]
 - Code : [1D] h + [47] h + [00] h
 - Setting "0" to the "bit7" of the printer status.
 - (3) [Print start setting (with JOB ID)]
 - Code : [1D] h + [47] h + [12] h + ID1 + ID2 + ID3 + ID4
 - Setting "1" to bit7 of printer status.
 - Setting JOB ID (4 bytes).
 - Holding the result of the logical sum from the start setting to the finish setting as a completion status.
 - (4) [Print finish setting (with JOB ID)]
 - Code : [1D] h + [47] h + [10] h
 - Setting "0" to the "bit7" of the printer status.
 - Transmitting the print finish notification with the following format.
 - [FF] h + [13] h + JOB ID (4 bytes) + finish status (1 byte) + backup (3 bytes)
2. Print start / finish setting (storing) (n = [20] h, [21] h, [30] h, [31] h)
 - When receiving the data, it will be stored as a printing image in the page buffer. Stored printing image will be printed by receiving "Print finish setting".
 - Although it will take a little from data receiving start to printing start, printing operation will become stable at the maximum printing speed. However, data of sheets with longer than 160mm printing length will be divided into parts for 160mm printing length and storing and printing operation will be repeated for 160mm printing length.
 - Printing will be halted once and will be restarted again for the next 160mm printing length data.
 - If a print stop error occurs during printing the data sandwiched between the print start setting and the print finish setting, the print data up to the print finish setting is disregarded.
 - (1) [Print start setting (storing) (without JOB ID)]
 - Code : [1D] h + [47] h + [21] h
 - Setting "1" to the "bit7" of the printer status and starting storing.
 - (2) [Print finish setting (storing) (without JOB ID)]
 - Code : [1D] h + [47] h + [20] h
 - Printing the image by the data stored in the page buffer and setting "0" to the "bit7" of the printer status.
 - Clearing the page buffer.

(3) [Print start setting (with JOB ID)]

Code : [1D] h + [47] h + [31] h + ID1 + ID2 + ID3 + ID4

Setting "1" to the "bit7" of the printer status and starting storing.

- Setting JOB ID (4 bytes).
- Holding the result of the logical sum from the start setting to the finish setting as a completion status.

(4) [Print finish setting (storing) (with JOB ID)]

Code : [1D] h + [47] h + [30] h

Printing the image by the data stored in the page buffer and setting "0" to the "bit7" of the printer status.

- Clearing the page buffer.
- Transmitting the print finish notification with the following format.
[FF] h + [13] h + JOB ID (4 bytes) + completion status (1 byte) + reserves (3 bytes)

[Notes]

- When using this command continuously with the USB interface, please use only "n = [31] h, [30] h" and be sure to send the next print start setting command after receiving the completion status.

51) [Print position selection for HRI characters] 《GS H n》 ◆

Code : [1D] h + [48] h + n ※ [00 ≤ n ≤ 03] h

Selecting the HRI characters printing position when printing barcodes.

- "n" has following meanings.

n (hex)	Printing position
00	No printing
01	Above the barcode
02	Under the barcode
03	Above and under the barcode

- HRI characters will be printed with the font selected by the command "[Font selection for HRI characters] 《GS f n》".
- Default value of "n" is "[00] h".

52) [Back feeding (mm unit)] 《GS J n》

Code : [1D] h + [4A] h + n ※ [00 ≤ n ≤ FF] h

Paper is fed to the reverse direction when this command is received.

- Specifying the feeding amount "n" by mm. In case of setting "[00] h", it will never be fed.
- Some deviation may occur due to taking the backlash.
- If there is print data in the print line buffer, back feeding will be started after printing them.
- Please DO NOT make the tip of paper exceed the back feeding limitation.
(Please refer to "2.4 Cutter specifications" in the Product Specifications.)

53) [Memory switch setting and set contents printing] 《GS M n d1 d2》

Code : [1D] h + [4D] h + n + d1 + d2

- ※ [00 ≤ n ≤ 01] h
- ※ [00 ≤ d1 ≤ FF] h
- ※ [00 ≤ d2 ≤ FF] h

Setting the memory switch MS1 and the memory switch MS2 and printing the set result.

- “n” has following meanings.

n (hex)	Function
00	Printing set contents of the memory switch MS1 and the memory switch MS2
01	Setting the memory switch

- When “n = 01” is set, “d1” and “d2” will become able to be set.
- When memory switches are set, set contents will be printed and then software reset operation will be performed automatically to enable the setting.
- When contents of each switch setting are printed, “upright + font A” characters will be printed temporarily for the specified character decoration and the setting will return to the original one after printing is finished. However, specified enhanced print characters will be printed as it is specified.
- “d1” specifies the memory switch MS1 setting by bytes and “d2” specifies the memory switch MS2 setting by bytes.

Each bit of “d1” and “d2” has following meanings.

d1 (Memory switch MS1)

bit	Function	Value	
		0	1
bit0	MS1-1	OFF	ON
bit1	MS1-2	OFF	ON
bit2	MS1-3	OFF	ON
bit3	MS1-4	OFF	ON
bit4	MS1-5	OFF	ON
bit5	MS1-6	OFF	ON
bit6	MS1-7	OFF	ON
bit7	MS1-8	OFF	ON

d2 (Memory switch MS2)

bit	Function	Value	
		0	1
bit0	MS2-1	OFF	ON
bit1	MS2-2	OFF	ON
bit2	MS2-3	OFF	ON
bit3	MS2-4	OFF	ON
bit4	MS2-5	OFF	ON
bit5	MS2-6	OFF	ON
bit6	MS2-7	OFF	ON
bit7	MS2-8	OFF	ON

- For details and precautions for each switch, please refer to “4.1.1 Switch setting” in the Product Specifications.

54) [Fixed bit image printing] 《GS P n》

Code : [1D] h + [50] h + n ※ [00 ≤ n ≤ 02] h

Printing the registered bit image printing data.

- Selecting one of three registered print patterns by specifying “00” to “02” for “n”.
- “n” has following meanings.

n (hex)	Printing pattern
00	Pattern 0
01	Pattern 1
02	Pattern 2

55) [Fixed bit image registration] 《GS T n》

Code : [1D] h + [54] h + n

Registering the print data of predetermined bit image or raster bit image.

- 3 patterns from pattern “0” to pattern “2” can be registered.
- Maximum 64K bytes bit image data or raster bit image data for each pattern can be registered. (Parts exceeding the maximum size will be discarded.)
- Please refer to the following formula regarding the data size to be registered.
Print area (mm) × vertically counted number of dots of bit image
× the number of bit images
- Registered data will not be erased either by turning the power OFF / ON, by executing the command “[Printer initialization] 《ESC @》”, by executing the command “[Software reset] 《DC1》” or executing reset by pressing the RESET button.
- “n” has following meanings.

n (hex)	Function
00	Starting pattern 0 registration
01	Starting pattern 1 registration
02	Starting pattern 2 registration
FF	Finishing registration

- When starting the bit image registration in the middle of a line, the whole line will be registered.
- When finishing the bit image registration in the middle of a line, the line will not be registered.

<Command sequence for registering pattern “0”>

When registering only bit image

“GS T [00] h + (Bit image data by ESC *) x “n lines” + “GS T [FF] h”

When registering only raster bit image

“GS T [00] h + (raster bit image by ESC b)” + “GS T [FF] h”

When registering mixed bit image / raster bit image

“GS T [00] h + (bit image data by ESC *) x “n lines” + (raster bit image by ESC b)
+ GS T [FF] h”

56) [Firmware download] 《GS d Dn》

Code : [1D] h + [64] h + Dn

Downloading the HEX file of the printer firmware, rewriting the firmware according to it and restarting the printer.

- “Dn” is the HEX code of firmware and it conforms to Intel HEX format.
- Please DO NOT rewrite the firmware carelessly.
- Please DO NOT make the power OFF while the firmware is being rewritten.
- Please rewrite the firmware at your own risk.

57) [Font selection for HRI characters] 《GS f n》 ◆

Code : [1D] h + [66] h + n ※ [00 ≤ n ≤ 01] h

Selecting the font for HRI characters when printing the barcode with HRI characters.

- "n" has following meanings.

n (hex)	Font
00	Font A
01	Font B

- Default value of "n" is " [00] h".

58) [Barcode height specifying] 《GS h n》 ◆

Code : [1D] h + [68] h + n ※ [01 ≤ n ≤ FF] h

Specifying the height of the barcode by dots.

- "n" indicates the number of dots counted vertically.
- Default value of "n" is " [A2] h (162 dots)".
- The bar height of GS1 DataBar will not become shorter than the minimum height described in the barcode specifications details.
- When the bar is put to the area with more than 2 columns by GS1 DataBar Stacked, GS1 DataBar Stacked Omnidirectional or GS1 DataBar Expanded Stacked, the height of one bar is 1/2 of the height specified by "n" . When 1/2 of the specified height is longer than the minimum height and the bar takes 2 columns, the barcode height becomes "specified value "n" + separator" height.

59) [Barcode print] 《GS k n Dn NUL》

* n = “ [08] h ~ [0E] h” is effective only with F/W version 1.10 or later.

Code : [1D] h + [6B] h + n + Dn + [00] h ※ [00 ≤ n ≤ 0E] h

Selecting the barcode symbology and printing the barcode.

- Beginning of the line shall be a next printing starting position.
- Selecting one of the following barcode symbology by the value of “n”.

n (hex)	Barcode symbology
00	UPC-A
01	UPC-E
02	JAN-13 (EAN-13)
03	JAN-8 (EAN-8)
04	CODE39
05	ITF
06	CODABAR (NW-7)
07	CODE128
08	CODE93
09	GS1 DataBar Omnidirectional
0A	GS1 DataBar Truncated
0B	GS1 DataBar Stacked
0C	GS1 DataBar Stacked Omnidirectional
0D	GS1 DataBar Limited
0E	GS1 DataBar Expanded (Stacked)

- “Dn” indicates the character code to be printed.
- When character code “Dn” is not printable character, the following data after “Dn” will be treated as normal print data.
- When selecting the barcode symbology which has the fixed number of printing characters, please be sure to make the number of characters equal to the number of printing characters.
- Barcode will not be printed when its horizontal data exceeds the length of a line except for GS1 DataBar Expanded.
- “ [00] h” at the end of this command can be changed to “ [FF] h” by the command “[Barcode termination changing] 《ESC RS c n》”.
- Please refer to “[Barcode detailed specifications]” from the next page for details of barcodes.

[Barcode detailed specifications]

Name	Numbers of characters	Types of characters	Remarks
UPC-A	12	Numbers (0~ 9)	- Check digit that has been calculated inside the printer will be automatically added to 12th digit. - Calculated value will be prioritized if the numerical value differs from 12th digit.
UPC-E	8	Numbers (0~ 9)	- The number of effective digits for UPC-A is 12 and that for UPC-E is 8. - Inside the printer, 11 digits excluding the check digit is compressed into 7 digits. - Check digit that has been calculated inside the printer will be automatically added to the 8th digit position. - Calculated value added to the 8th digit is prioritized if it is different from the value of the 12th digit before compressed.
JAN-13 (EAN-13)	13	Numbers (0~ 9)	- Check digit that has been calculated inside the printer will be automatically added to 13th digit. - Calculated value will be prioritized if the numerical value differs from 13th digit.
JAN-8 (EAN-8)	8	Numbers (0~ 9)	- Check digit that has been calculated inside the printer will be automatically added to 8th digit. - Calculated value will be prioritized if the numerical value differs from 8th digit.
CODE39	Variable length	Numbers (0~ 9) Alphabets (A~ Z) Marks (\$%*+-./:SPACE) Start/Stop Code (*)	Please be sure to enter start / stop code “*”.
ITF	Even numbers	Numbers : 0~ 9	Odd numbers will not be printed.
CODABAR (NW-7)	Variable length	Numbers (0~ 9) Marks (\$+-./:SPACE) Start/Stop Code (ABCD) (abcd)	- Please be sure to enter “start / stop code”. - Although it can be printed out without “start / stop code”, reading it by scanner, etc. cannot be performed.
CODE93	Variable length	Numbers (0~ 9) Alphabet (A~ Z) (a~ z) Control character ([00] h ~ [1F] h) ([7F] h)	”□” is automatically added to before and after HRI characters. - HRI character of the control character is printed by the combination of “■” and one alphabetic character.*1
CODE128	Please refer to “CODE128 in [Barcode detailed specifications]”.		
GS1 DataBar	Please refer to “GS1 DataBar in [Barcode detailed specifications]”.		

*1: CODE93 control characters' notation by HRI characters

ASCII	Hexadecimal	HRI characters	ASCII	Hexadecimal	HRI characters
NULL	00	■U	DLE	10	■P
SOH	01	■A	DC1	11	■Q
STX	02	■B	DC2	12	■R
ETX	03	■C	DC3	13	■S
EOT	04	■D	DC4	14	■T
ENQ	05	■E	NAK	15	■U
ACK	06	■F	SYN	16	■V
BEL	07	■G	ETB	17	■W
BS	08	■H	CAN	18	■X
HT	09	■I	EM	19	■Y
LF	0A	■J	SUB	1A	■Z
VT	0B	■K	ESC	1B	■A
FF	0C	■L	FS	1C	■B
CR	0D	■M	GS	1D	■C
SO	0E	■N	RS	1E	■D
SI	0F	■O	US	1F	■E
			DEL	7F	■T

[Barcode detailed specifications]

• CODE128

1. Overview

ASCII 128 characters (numbers, uppercase / lowercase alphabetical letters, symbols, control characters) can be represented and one bar pattern can represent numbers with two digits. The number of characters is variable and the stop character and the check digit will be added automatically.

2. Types of characters

- Code set A : ASCII characters in " [00] h ~ [5F] h" can be represented.
- Code set B : ASCII characters in " [20] h ~ [7F] h" can be represented.
- Code set C : Numbers with 2 digits (00 ~ 99) can be represented by one bar pattern.
- Special characters

(1) Start character / code definition character (CODE A, CODE B and CODE C)

The head of the barcode data has to be this character.

You can change the code set from one to the other in the middle of the data.

(2) Shift character (SHIFT)

Code set A: One character immediately after SHIFT is treated as a character from code set B.

Code set B: One character immediately after SHIFT is treated as a character from code set A.

Code set C: Unavailable

(3) Function characters (FNC1, FNC2, FNC3 and FNC4)

Uses of these characters depend on applications.

"FNC1" can be used only for code set C.

3. Detailed specifications

- Command execution is halted if the barcode data does not start with a "start character".
- Command execution is halted if the combination of "{" and the next character does not fall under special characters.
- To use "{" as a character, please input ""{"([7B]h + [7B]h)".
- Although the barcode will be printed when there are any characters which is not included in the selected code set, it cannot be read by a scanner, etc.
- To use " [00] h" as a code, please change the barcode termination character to " [FF] h" with the command "[Barcode termination changing] 《ESC RS c n》".
- Specifications of HRI characters of control / special characters:

(1) Control characters ([00] h ~ [1F] h, [7F] h)

A space is displayed.

(2) Start character / code definition character (CODE A, CODE B and CODE C)

Nothing is printed.

(3) Shift character (SHIFT)

Nothing is printed.

(4) Function characters (FNC1, FNC2, FNC3 and FNC4)

A space character is printed.

4. Character set table

Special characters

	ASCII	Hexadecimal
CODE A	{A	[7B] h + [41] h
CODE B	{B	[7B] h + [42] h
CODE C	{C	[7B] h + [43] h
SHIFT	{S	[7B] h + [53] h
FNC 1	{1	[7B] h + [31] h
FNC 2	{2	[7B] h + [32] h
FNC 3	{3	[7B] h + [33] h
FNC 4	{4	[7B] h + [34] h

[Barcode detailed specifications]

Code set A

HEX	0	1	2	3	4	5
0	NULL	DLE	SP	0	@	P
1	SOH	DC1	!	1	A	Q
2	STX	DC2	"	2	B	R
3	ETX	DC3	#	3	C	S
4	EOT	DC4	\$	4	D	T
5	ENQ	NAK	%	5	E	U
6	ACK	SYN	&	6	F	V
7	BEL	ETB	'	7	G	W
8	BS	CAN	(	8	H	X
9	HT	EM	)	9	I	Y
A	LF	SUB	*	:	J	Z
B	VT	ESC	+	;	K	[
C	FF	FS	,	<	L	¥
D	CR	GS	-	=	M	]
E	SO	RS	.	>	N	^
F	SI	US	/	?	O	_

Code set B

HEX	2	3	4	5	6	7
0	SP	0	@	P	`	p
1	!	1	A	Q	a	q
2	"	2	B	R	b	r
3	#	3	C	S	c	s
4	\$	4	D	T	d	t
5	%	5	E	U	e	u
6	&	6	F	V	f	v
7	'	7	G	W	g	w
8	(	8	H	X	h	x
9	)	9	I	Y	i	y
A	*	:	J	Z	j	z
B	+	;	K	[	k	{
C	,	<	L	¥	l	
D	-	=	M	]	m	}
E	.	>	N	^	n	~
F	/	?	O	_	o	DEL

Code set C

HEX	0	1	2	3	4	5	6
0	00	16	32	48	64	80	96
1	01	17	33	49	65	81	97
2	02	18	34	50	66	82	98
3	03	19	35	51	67	83	99
4	04	20	36	52	68	84	
5	05	21	37	53	69	85	
6	06	22	38	54	70	86	
7	07	23	39	55	71	87	
8	08	24	40	56	72	88	
9	09	25	41	57	73	89	
A	10	26	42	58	74	90	
B	11	27	43	59	75	91	
C	12	28	44	60	76	92	
D	13	29	45	61	77	93	
E	14	30	46	62	78	94	
F	15	31	47	63	79	95	

[Barcode details specifications]

• GS1 DataBar

Name	Numbers of characters	Types of characters	Minimum height	Remarks
GS1 DataBar Omnidirectional	Not more than 13	Numbers (0~ 9)	33 modules • In case of horizontal size = 3 $33 \times 3 = 99$ dots	• Check digit calculated in the printer will be automatically added to 14th digit. • When the number of input characters is less than 13, it is automatically filled with 0s from the first digit in order to make it 13 characters. • HRI characters are automatically prefixed with "(01)".
GS1 DataBar Truncated	Not more than 13	Numbers (0~ 9)	13 modules • In case of horizontal size = 3 $13 \times 3 = 39$ dots	• Check digit calculated in the printer will be automatically added to 14th digit. • When the number of input characters is less than 13, it is automatically filled with 0s from the first digit in order to make it 13 characters. • HRI characters are automatically prefixed with "(01)".
GS1 DataBar Stacked	Not more than 13	Numbers (0~ 9)	Top : 5 modules SPR : 1 module Bottom : 7 modules • In case of horizontal size = 3 Top : $5 \times 3 = 15$ dots SPR : $1 \times 3 = 3$ dots Bottom : $7 \times 3 = 21$ dots	• Check digit calculated in the printer will be automatically added to 14th digit. • When the number of input characters is less than 13, it is automatically filled with 0s from the first digit in order to make it 13 characters. • HRI characters are automatically prefixed with "(01)".
GS1 DataBar Stacked Omnidirectional	Not more than 13	Numbers (0~ 9)	Top : 33 modules SPR : 3 modules Bottom : 33 modules • In case of horizontal size = 3 Top : $33 \times 3 = 99$ dots SPR : $3 \times 3 = 9$ dots Bottom : $33 \times 3 = 99$ dots	• Check digit calculated in the printer will be automatically added to 14th digit. • When the number of input characters is less than 13, it is automatically filled with 0s from the first digit in order to make it 13 characters. • HRI characters are automatically prefixed with "(01)".
GS1 DataBar Limited	Not more than 13	Numbers (0~ 9)	10 modules • In case of horizontal size = 3 $10 \times 3 = 30$ dots	• Check digit calculated in the printer will be automatically added to 14th digit. • When the number of input characters is less than 13, it is automatically filled with 0s from the first digit in order to make it 13 characters. • HRI characters are automatically prefixed with "(01)".

				The number that can be expressed is from 0 to 1,999,999,999,999.
GS1 DataBar Expanded (Stacked)	Variable length (4~74)	ISO646 : - Numbers (0~9) - Alphabets (A~Z) (a~z) - Space - FNC1 20 marks (!, ", %, &, ', (,), *, +, -, ., /, :, ;, <, >, =, ?, _)	Per bar : 34 modules SPR : 3 modules In case of horizontal size = 3 1) In case of one bar $34 \times 3 = 102$ dots 2) In case of 3 bars $34 \times 3 + 3 \times 3 + 34 \times 3 + 3 \times 3 + 34 \times 3 = 324$ dots	- Up to 74 numbers or up to 41 letters can be encoded at maximum. 4 digits are required at minimum. - If the horizontal size exceeds the print area, it will be printed in the area with multistage bars automatically. - When print area is less than 2 segments of the barcode, it will not be printed. - HRI characters will not represent AI expression. - When AI expression is required with HRI characters, please create a character string with AI expression on the host side and print it before or after the barcode.

[Notes]

- Separator is described as "SPR".
- When the bar is put to the area with more than 2 stages by GS1 DataBar Stacked, GS1 DataBar Stacked Omnidirectional or GS1 DataBar Expanded Stacked, the height of 1 stage will be 1/2 of the height specified by the command "[Barcode height specifying] 《GS h n》". When 1/2 of the specified height is more than the minimum height and the bar is put into the area with 2 stages, "specified value + separator height" becomes the barcode height.

[Barcode details specifications]

GS1 DataBar Expanded character set table

HEX	2	3	4	5	6	7
0	SP	0		P		p
1	!	1	A	Q	a	q
2	"	2	B	R	b	r
3	FNC1	3	C	S	c	s
4		4	D	T	d	t
5	%	5	E	U	e	u
6	&	6	F	V	f	v
7	'	7	G	W	g	w
8	(	8	H	X	h	x
9	)	9	I	Y	i	y
A	*	:	J	Z	j	z
B	+	;	K		k	
C	,	<	L		l	
D	-	=	M		m	
E	.	>	N		n	
F	/	?	O	=	o	

60) [Automatic printer status transmission] 《GS v NUL》

Code : [1D] h + [76] h + [00] h

Transmitting the printer status automatically when it is changed.

- Transmitted status is one byte data. Please refer to “[4.2 Processing errors 1) Details of error detection]” in the Product Specifications. Furthermore, please refer to “[3.2 6) Transmitted data]” in the Product Specifications for the transmission format.
- Once the setting is done, it will be effective until executing the command “[Software reset] 《DC1》”, executing reset by pressing the RESET button or turning the power OFF.
- This command is stored in the internal receiving input buffer and is executed sequentially.
- This command is effective only with serial interface.
- Please refer to “[4.1.1 Switch setting]” in the Product Specifications when using XON / XOFF control under the interface set to serial.

61) [Barcode horizontal size specifying] 《GS w n》 ◆

Code : [1D] h + [77] h + n

※ [02≤n≤04] h

Specifying the barcode horizontal size.

- Default value of “n” is “ [03] h”.

62) [Print density setting] 《GS ~ n》

Code : [1D] h + [7E] h + n

※ [41≤n≤A5] h

Setting the print density in the range from 65% to 165%.

- Although the range by “n” starts at “ [41] h (65%)” and ends at “ [A5] h (165%)”, please set it from “ [41] h (65%)” to “ [87] h (135%)” in actual use.
- Default value of “n” is “ [64] h” (100%).
- When using this command during the command “[Print start / finish setting] 《GS G n》” execution, the setting will be disregarded.

63) [Maximum printing speed setting] 《GS S n》

Code : [1D] h + [53] h + n

※ [00≤n≤04] h

Setting the maximum printing speed during printing after storing.

- “n” has following meanings.

n (hex)	Maximum printing speed
00	Max. 150mm / sec
01	
02	
02	Max. 125mm / sec

n (hex)	Maximum printing speed
03	Max. 100mm / sec
04	Max. 75mm / sec

- Default value of “n” is “ [00] h”.

64) [Specifying Kanji print modes all at once] 《FS ! n》 ◆

Code : [1C] h + [21] h + n

※ [00 ≤ n ≤ FF] h

Specifying all Kanji print modes at one time.

- “n” has following meanings.

bit	Function	Value	
		0	1
bit0	Undefined		
bit1	Undefined		
bit2	Double width	Releasing	Specifying
bit3	Double height	Releasing	Specifying
bit4	Undefined		
bit5	Undefined		
bit6	Undefined		
bit7	Underline	Releasing	Specifying

- If double height and double width are set at the same time, character with double-height and double-width will be formed.
- When underline is specified, all printed characters will be underlined except for 90° clock-wise rotated characters and spaces created by the command “[Horizontal tab] 《HT》”.
- Underline width is determined by the value set in the command “[Specifying / releasing underline to Kanji] 《FS - n》”.
- Default value is “1 dot” width.
- Characters mixed with different sizes of normal size, double height, double width or double height and double width can be printed.
- Combined printing with ANK characters is also available.
- Default value of “n” is “ [00] h”.

65) [Kanji mode specifying] 《FS &》 ◆

Code : [1C] h + [26] h

Specifying the Kanji mode.

- It will be ineffective when Kanji code system is Shift JIS.
- Default status is the status with Kanji mode released.

66) [Specifying / releasing underline to Kanji] 《FS - n》 ◆

Code : [1C] h + [2D] h + n

※ [00 ≤ n ≤ 02] h

Specifying / Releasing underline to Kanji.

- All printed Kanji characters will be underlined except for the 90° clockwise rotated characters and spaces created by the command “[Horizontal tab] 《HT》”.
- This command is not effective when Kanji mode is released.
- “n” has following meanings.

n (hex)	Function
00	Releasing underline to Kanji
01	Specifying one dot height underline to Kanji
02	Specifying two dots height underline to Kanji

- Default value of “n” is “ [00] h”.

67) [Kanji mode releasing] 《FS .》

Code : [1C] h + [2E] h

Releasing the Kanji mode.

- It will be ineffective when Kanji code system is Shift JIS.
- Default status is the status with Kanji mode released.

68) [Extended characters definition] 《FS 2 a1 a2 Dn》

Code : [1C] h + [32] h + a1 + a2 + Dn

JIS code system

※ [a1=77] h

※ [21≤a2≤7E] h

Shift JIS code system

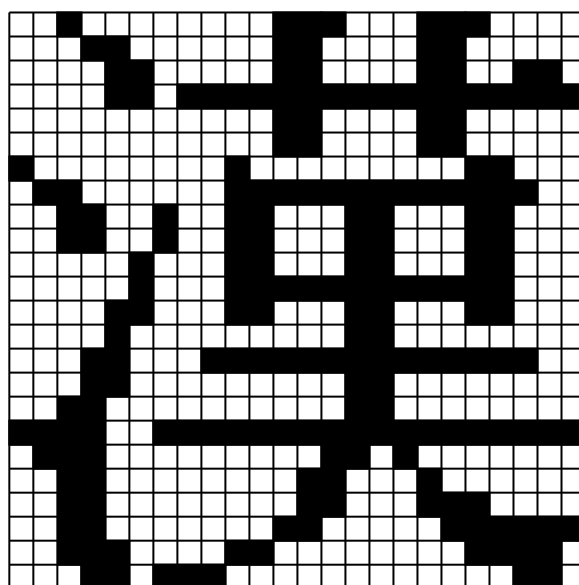
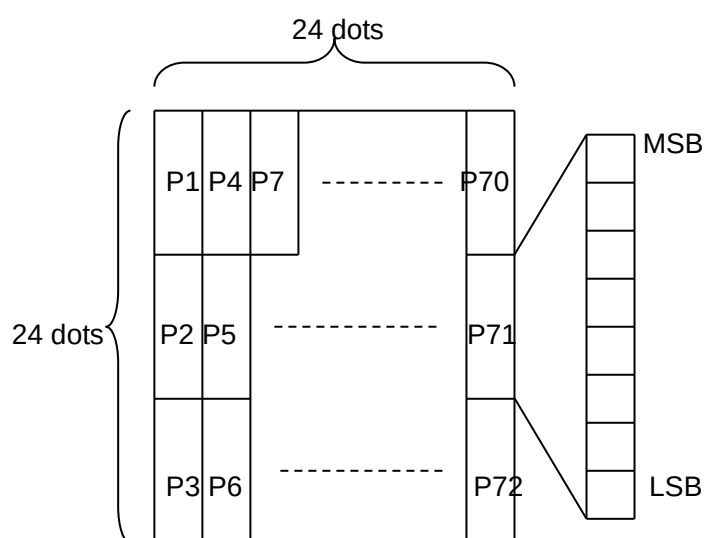
※ [a1=EC] h

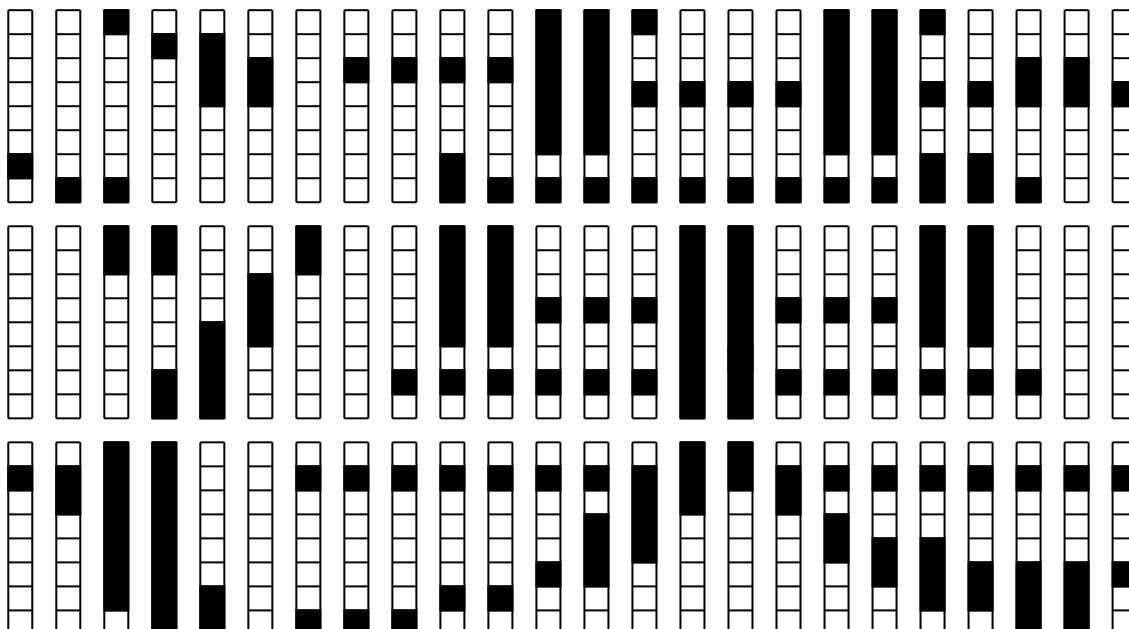
※ [40≤a2≤7E, 80≤a2≤9E] h

Defining extended Kanji characters.

- 94 characters can be defined.
- “Dn” is data to be defined. Data size will be 72 bytes (3 bytes (vertical) × 24 dots (horizontal)).
- Default status has “spaces” filled in all cells.
- Once it is defined by this command, it is effective until executing the command “[Software reset] 《DC1》”, executing reset by pressing the RESET button or turning OFF the power.
- Only specified area can be redefined.

<Reference>





P1= [02] h,P4= [01] h,P7= [81] h,P10= [40] h,P13= [70] h,P16= [30] h, ...
P2= [00] h,P5= [00] h,P8= [C0] h,P11= [C3] h,P14= [0F] h,P17= [38] h, ...
P3= [40] h,P6= [60] h,P9= [FE] h,P12= [FF] h,P15= [03] h,P18= [00] h, ...

69) [Kanji code system selection] 《FS C n》 ◆

Code : [1C] h + [43] h + n

※ [00 ≤ n ≤ 01] h

Selecting Kanji code system.

- “n” has following meanings.

n (hex)	Code
00	JIS code
01	Shift JIS code

- Default status is a code specified by “4.1.1. Switch setting” described in the Product Specifications.

70) [Kanji space amount specifying] 《FS S n1 n2》 ◆

Code : [1C] h + [53] h + n1 + n2

※ [00 ≤ n1 ≤ 20] h

※ [00 ≤ n2 ≤ 20] h

Specifying the amount of spaces in the right and the left of Kanji characters by the unit of dots.

- “n1” specifies the left spaces amount. Default value of “n1” is “ [00] h”.
- “n2” specifies the right spaces amount. Default value of “n2” is “ [00] h”.
- When double width mode is selected, width of spaces will be 2 times wide.

71) [Double-height and double-width Kanji size specifying / releasing] 《FS W n》 ◆

Code : [1C] h + [57] h + n

※ [00 ≤ n ≤ FF] h

Specifying / Releasing double-height and double-width Kanji characters.

- Only LSB (least significant bit) (“bit0”) is available for “n”.
- Ways of controlling by the LSB “bit0” are described in the table below.

bit0	Function
0	Releasing double-height and double-width
1	Specifying double-height and double-width

- Default value of “n” is “ [00] h”.

72) [USB serial No. setting] 《GS U Dn》

Code : [1D] h + [55] h + Dn

Setting the USB serial number.

- "Dn" indicates USB serial No. data.
- It has fixed 8 bytes length and available characters are in " "([20] h), "-" ([2D] h), "/" ([2F] h), [30] h to [7A] h.
- In case of " "Dn" = " [00] h (1 byte)", default value of USB serial No. will be set.
- USB serial No. by this setting will become effective when activating the power for the next time.
- Default value is "without USB serial No."
- USB serial No. can be checked by the self-diagnostic printing result. Please refer to "4.1.4 Self-diagnostic printing" in the Product Specifications.

73) [Page layout setting] 《ESC T Dt》

* This command is one of commands related to the page layout.

Code : [1B] h + [54] h + Dt ※ [00 ≤ Dt ≤ 03] h

Setting the page direction.

- "Dt" has following meanings.

Dt (hex)	Mode
00	Portrait mode (not rotated)
01	Landscape mode (90° rotated)
02	Portrait mode (180° rotated)
03	Landscape mode (270° rotated)

- Default value of "Dt" is " [00] h". [Portrait mode (not rotated)]
- This command initializes the page length setting and the print area setting.
- If barcodes are printed on the landscape layout page, printing may be dragged and it may not be read. In this case, it is necessary to reduce the printing speed by the command "[Maximum printing speed setting] 《GS S n》" or shorten the length of the barcode.

74) [Page length setting] 《ESC I DI》

* This command is one of commands related to the page layout.

Code : [1B] h + [6C] h + DI

Setting the page length.

- Paper feeding length during the page print can be changed by changing the page length.
- Specifying "DI" by "mm" for the page length.
- Page length is the length to Y direction for the portrait setting and the length to X direction for the landscape setting.
- It will initialize the page layout print area setting.
- Please refer to the next page (in case of portrait mode) and after the next page (in case of landscape mode).

75) [Page layout print area setting] 《ESC W DI Dt Dw Dh》

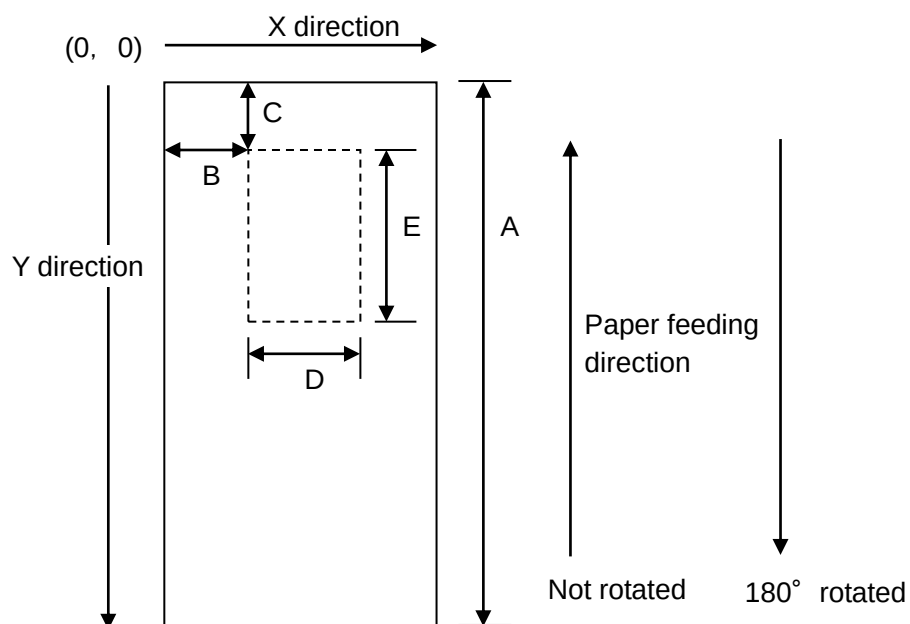
* This command is one of commands related to the page layout.

Code : [1B] h + [57] h + DI + Dt + Dw + Dh

Setting the print area of page layout.

- Specifying the left margin by "DI" and the upper margin by "Dt" by "mm".
- Specifying the X print area by "Dw" and the Y print area by "Dh" by "mm".
- Only effective at the head of a line.
- Please refer to the next page (in case of portrait mode) and after the next page (in case of landscape mode).

<In case of the portrait page>

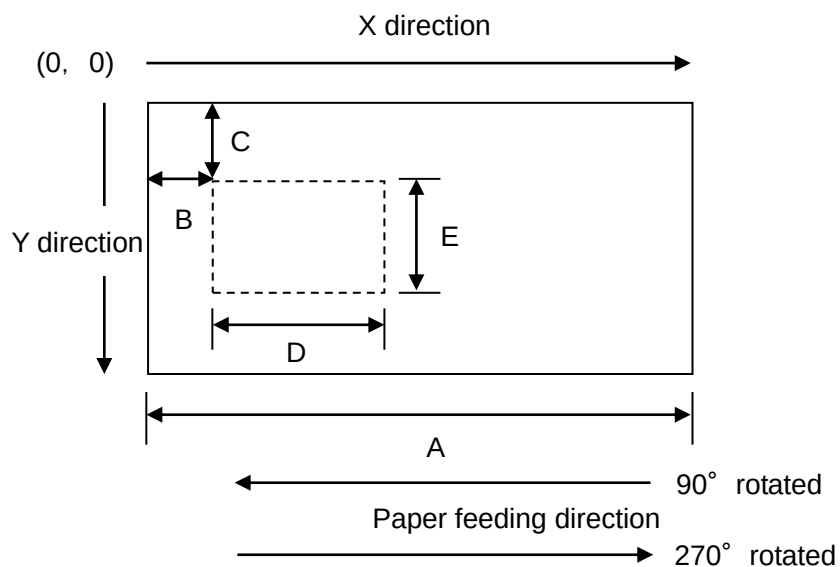


Mark	Descriptions	Minimum	Maximum	Default setting
A	Page length	3mm	160mm	160mm
B	Left margin	0mm	69mm	4mm
C	Upper margin	0mm	157mm	0mm
D	X print area	3mm	72mm	72mm
E	Y print area	3mm	160mm	160mm

[Notes]

- Taking the left margin and the right margin both for longer than 3mm is recommended. Please refer to “2.3 Print area” in the Product Specifications.

<In case of the landscape page>



Mark	Descriptions	Minimum	Maximum	Default setting
A	Page length	3mm	160mm	160mm
B	Left margin	0mm	157mm	0mm
C	Upper margin	0mm	69mm	4mm
D	X print area	3mm	160mm	160mm
E	Y print area	3mm	72mm	72mm

[Notes]

- Taking the left merging and the right margin both for longer than 3mm is recommended. Please refer to “2.3 Print area” in the Product Specifications.

76) [Page print and clear] 《FF》

* This command is one of commands related to the page print.

Code : [0C] h

Clearing the page buffer after printing contents in the page buffer.

- If this command is used during “Print start / finish setting (storing)” of the command “[Print start / finish setting] 《GS G n》”, it will be disregarded.

77) [Page print] 《ESC FF n》

* This command is one of commands related to the page print.

Code : [1B] h + [0C] h + n ※ [00≤n≤01] h

Printing the contents in the page buffer.

- “n” has following meanings.

n (hex)	Mode
00	Page print
01	Inverted page print

- Default value of “n” is “ [00] h (Page print)”.
- Inverted page print prints 180° rotated page.
- If this command is used during “Print start / finish setting (storing)” of the command “[Print start / finish setting] 《GS G n》”, it will be disregarded.

78) [Page buffer clear] 《CAN》

* This command is one of commands related to the page buffer clearing.

Code : [18] h

Clearing the page buffer.

79) [Page buffer area clear] 《ESC CAN Dx1 Dx2 Dy1 Dy2 Dw1 Dw2 Dh1 Dh2》

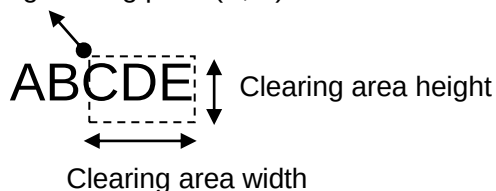
* This command is one of commands related to page buffer clearing.

Code : [1B] h + [18] h + Dx1 + Dx2 + Dy1 + Dy2 + Dw1 + Dw2 + Dh1 + Dh2

Clearing the specified area in the page buffer.

- Specifying dots for “clearing starting point” by (Dx2×256+Dx1) for X-axis coordinate, (Dy2×256+Dy1) for Y-axis coordinate.
- Specifying dots for “clearing area” by (Dw2×256+Dw1) for width, (Dh2×256+Dh1) for height.
- Please specify the total value of the clearing starting coordinates (X: Y) and the clearing area (width and height) within 65535.
(X-axis coordinate + width ≤ [FFFF] h: Y-axis coordinate + height ≤ [FFFF] h).
- The origin shall be upper-left corner of the print area.
- When “clearing starting point” is out of the print area, it will forcibly be shifted into the print area after clearing excessive coordinates to “0”.

Clearing starting point (X, Y)



80) [Page buffer layout starting / finishing setting] 《ESC L n》

Executing page buffer editing by one of following 2 ways.

- If this command is used during “Print start / finish setting (storing)” of the command “[Print start/finish setting] 《GS G n》”, it will be disregarded.

1. Page buffer layout starting / finishing setting (n= [00] h, [01] h)

(1) [Page buffer layout start setting]

Code : [1B] h + [4C] h + [01] h

Starting page buffer layout.

- Executing layout in line by line.
- Initializing layout position to upper-left of print area.

(2) [Page buffer layout finish setting]

Code : [1B] h + [4C] h + [00] h

Finishing page buffer layout.

2. Page buffer layout starting / finishing setting with specified coordinates

(n= [10] h, [11] h)

(1) [Setting the start of page buffer layout with specified coordinates]

Code : [1B] h + [4C] h + [11] h + Dx1 + Dx2 + Dy1 + Dy2 + Dt + Dv

Starting the page buffer layout from the set coordinates position.

- Setting dots for starting point with (Dx2×256+Dx1) for X-axis coordinate and with (Dy2×256+Dy1) for Y-axis coordinate.
- “Dt” sets rotation and “Dv” sets overlay.
- The origin shall be the upper-left corner of the print area.
- When “layout starting point” is out of the print area, it will forcibly shift into the print area after clearing excessive coordinates to “0”.
- Executing layout object by object (character string, barcode, bit image and raster bit image). Excessive part from the print area will be discarded.
- Only the flag changes by the command “[Position alignment] 《ESC a n》”.
- The command “[Fixed bit image registration] 《GS T n》” will be disregarded.
- “Dt” has following meanings.

Dt (hex)	Mode
00	Not rotated
01	Rotation of 90° clockwise

- “Dv” has following meanings.

Dv (hex)	Mode
00	Overwriting background
01	Overlaying background

(2) [Setting the finish of page buffer layout with specified coordinates]

Code : [1B] h + [4C] h + [10] h

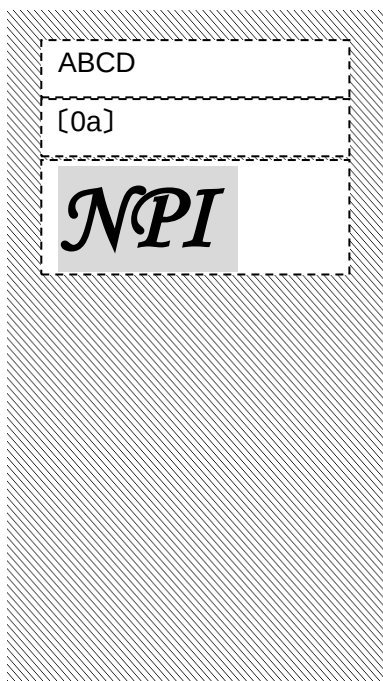
Finishing the page buffer layout with specified coordinates.

[Notes]

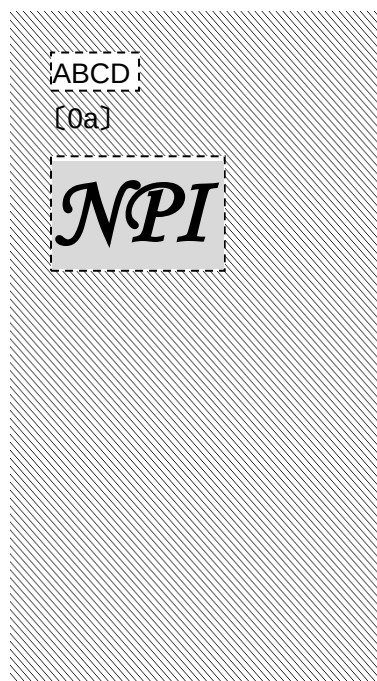
- The page print sample is shown on the next page.

- Page print sample :
 - (1) Page layout setting (setting the page direction)
 - (2) Page layout print area setting
 - Left margin / Upper margin=0mm
 - X print area=80mm
 - Y print area=160mm
 - (3) Starting page buffer layout
 - (4) Background image (bit image)
 - (5) Finishing page buffer layout
 - (6) Page layout print area setting (e.g. 1 / e.g. 2)
 - Left margin / Upper margin=4mm
 - X print area=72mm
 - Y print area=152mm
 - (7) Starting page buffer layout (e.g. 1) / Starting page buffer layout with specified coordinates (e.g. 2)
 - (8) Character string "ABCD" + [0a] h
 - (9) [0a] h
 - (10) LOGO image (bit image : NPI)
 - (11) Finishing page buffer layout (e.g. 1) / Finishing page buffer layout with specified coordinates set page buffer layout finish (e.g. 2)

e.g. 1



e. g. 2



- It will be overwritten to the print area because the layout is by line.

- It will be overwritten or overlaid to the height and the width of the object because it will be overlaid by object.

[Notes]

- Data which is laid out to the area exceeding Y printing area will not be printed.

3. Character code table

3.1 Domestic character code table

	HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
HEX	BIN	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	NUL		SP	0	@	P	`	p	_	⌞	SP	-	タ	ミ	=	X
1	0001		DC1	!	1	A	Q	a	q	■	〒	。	ア	チ	ム	ト	円
2	0010			”	2	B	R	b	r	■	†	「	イ	ツ	メ	キ	年
3	0011		DC3	#	3	C	S	c	s	■	ト	」	ウ	テ	モ	†	月
4	0100			\$	4	D	T	d	t	■	〒	、	エ	ト	ヤ	▲	日
5	0101			%	5	E	U	e	u	■	—	・	オ	ナ	1	▲	時
6	0110			&	6	F	V	f	v	■		ヲ	カ	ニ	ヨ	▼	分
7	0111			'	7	G	W	g	w	■		ア	キ	ヌ	ラ	▼	秒
8	1000			(	8	H	X	h	x		「	イ	ク	ネ	リ	♠	〒
9	1001	HT		)	9	I	Y	i	y		」	ウ	ケ	ノ	ル	♥	市
A	1010	LF		*	:	J	Z	j	z		「	エ	コ	ハ	レ	♦	区
B	1011		ESC	+	;	K	[	k	{		」	オ	サ	ヒ	□	♣	町
C	1100	FF	FS	,	<	L	¥	l	!	■	「	ヤ	シ	フ	ワ	●	村
D	1101	CR	GS	-	=	M	]	m	}	■	、	ユ	ズ	ハ	ソ	0	人
E	1110		RS	.	>	N	^	n	~	■	、	ヨ	セ	ホ	”	/	■
F	1111			/	?	0	_	o	SP	+	、	ツ	ソ	マ	°	\	SP

[Notes]

- [SP] indicates space.
- [CR] will be disregarded.
- Printer operation cannot be guaranteed if a blank code in the controlling codes having the equal to or smaller than " [1F] h" number (indicated by the combination of two hexadecimal numbers) or an undefined code is transmitted to the printer.
- Please DO NOT transmit codes that are not described in this document to the printer.
- This code table indicates simplified symbols and does not represent the actual printing result by the printer.

3.2 Overseas character code table

	HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
HEX	BIN	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
<u>0</u>	<u>0000</u>	NUL		SP	0	@	P	`	p	Ç	É	á	÷	Ł	⌌	α	≡
<u>1</u>	<u>0001</u>		DC1	!	1	A	Q	a	q	ü	æ	í	⌘	⊥	⌒	β	±
<u>2</u>	<u>0010</u>			"	2	B	R	b	r	é	Æ	ó	⌘	⌒	π	Γ	≥
<u>3</u>	<u>0011</u>		DC3	#	3	C	S	c	s	â	ô	ú		⌒	⌌	π	≤
<u>4</u>	<u>0100</u>			\$	4	D	T	d	t	ä	ö	ñ	⌒	⌒	⌒	Σ	↑
<u>5</u>	<u>0101</u>			%	5	E	U	e	u	à	ò	Ñ	⌒	⌒	⌒	σ	↓
<u>6</u>	<u>0110</u>			&	6	F	V	f	v	ã	û	ä	⌒	⌒	⌒	μ	÷
<u>7</u>	<u>0111</u>			'	7	G	W	g	w	ç	ù	ó	⌒	⌒	⌒	τ	≈
<u>8</u>	<u>1000</u>			(	8	H	X	h	x	ê	ÿ	¿	⌒	⌒	⌒	°	
<u>9</u>	<u>1001</u>	HT		)	9	I	Y	i	y	ë	Ö	⌒	⌒	⌒	⌒	•	
<u>A</u>	<u>1010</u>	LF		*	:	J	Z	j	z	è	Ü	⌒	⌒	⌒	⌒	•	
<u>B</u>	<u>1011</u>		ESC	+	:	K	[	k	{	ï	ø	½	⌒	⌒	⌒	⌒	
<u>C</u>	<u>1100</u>	<u>FF</u>	<u>FS</u>	,	<	L	¥			î	£	¼	⌒	⌒	⌒	∞	n
<u>D</u>	<u>1101</u>	<u>CR</u>	<u>GS</u>	-	=	M	]	m	}	ì	¥	í	⌒	⌒	⌒	φ	²
<u>E</u>	<u>1110</u>		RS	.	>	N	^	n	~	Ä	ß	«	⌒	⌒	⌒	€	■
<u>F</u>	<u>1111</u>			/	?	O	_	o	SP	À	f	»	⌒	⌒	⌒	∩	SP

[Notes]

- [SP] indicates space.
- [CR] will be disregarded.
- Printer operation cannot be guaranteed if a blank code in the controlling codes having the equal to or smaller than " [1F] h" number (indicated by the combination of two hexadecimal numbers) or an undefined code is transmitted to the printer.
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- This code table indicates simplified symbols and does not represent the actual printing result by the printer.

3.3 CODE PAGE 858

	HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
HEX	BIN	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	NUL		SP	0	@	P	`	p	Ç	É	á	■	L	Œ	Ó	-
1	0001		DC1	!	1	A	Q	a	q	ü	æ	í	■	⊥	Ð	ß	±
2	0010			"	2	B	R	b	r	é	Æ	ó	■	⊥	Ê	Ô	=
3	0011		DC3	#	3	C	S	c	s	â	ô	ú			Ë	Ò	¾
4	0100			\$	4	D	T	d	t	ä	ö	ñ		—	È	Õ	¶
5	0101			%	5	E	U	e	u	à	ò	Ñ	Á	+	€	Ö	§
6	0110			&	6	F	V	f	v	â	û	ä	Â	ã	Í	μ	÷
7	0111			'	7	G	W	g	w	ç	ù	ó	À	Ã	Î	þ	,
8	1000			(	8	H	X	h	x	ê	ÿ	¿	©	ℓ	ï	ð	°
9	1001	HT		)	9	I	Y	i	y	ë	ö	®	¶	¶	⌋	Ú	''
A	1010	LF		*	:	J	Z	j	z	è	Ü	¬		⌋	⌋	Û	-
B	1011		ESC	+	;	K	[	k	{	ï	ø	½	¶	¶	■	Ü	¹
C	1100	FF	FS	,	<	L	\			î	£	¼	¶	¶	■	Ý	³
D	1101	CR	GS	-	=	M	]	m	}	ì	Ø	í	ø	=	!	Ÿ	²
E	1110		RS	.	>	N	^	n	~	Ä	×	«	¥	¶	Ï	—	■
F	1111			/	?	O	_	o	SP	Å	f	»	¶	¶	■	'	SP

[Notes]

- [SP] indicates space.
- [CR] will be disregarded.
- Printer operation cannot be guaranteed if a blank code in the controlling codes having the equal to or smaller than " [1F] h" number (indicated by the combination of two hexadecimal numbers) or an undefined code is transmitted to the printer.
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3.4 International character code table

n	Character set	23h	24h	40h	5Bh	5Ch	5Dh	5Eh	60h	7Bh	7Ch	7Dh	7Eh
00h	U.S.A.	#	\$	@	[	\	]	^	`	{		}	~
01h	France	#	\$	à	°	ç	§	^	`	é	ù	è	¨
02h	Germany	#	\$	§	Ä	Ö	Ü	^	`	ä	ö	ü	ß
03h	U.K.	£	\$	@	[	\	]	^	`	{		}	~
04h	Denmark I	#	\$	@	Æ	Ø	Å	^	`	æ	ø	å	~
05h	Sweden	#	¤	É	Ä	Ö	Å	Ü	é	ä	ö	å	ü
06h	Italy	#	\$	@	°	\	é	^	ù	à	ò	è	ì
07h	Spain	¤	\$	@	í	ñ	¿	^	`	¨	ñ	}	~
08h	Japan	#	\$	@	[	¥	]	^	`	{		}	~
09h	Norway	#	¤	É	Æ	Ø	Å	Ü	é	æ	ø	å	ü
0Ah	Denmark II	#	\$	É	Æ	Ø	Å	Ü	é	æ	ø	å	ü

[Notes]

- This code table indicates simplified symbols and does not represent the actual printing result by the printer.

3.5 CODE PAGE 1250

	HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
HEX	BIN	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	NUL		SP	0	@	P	`	p	€	SP	SP	°	Ř	Đ	ř	ď
<u>1</u>	<u>0001</u>		DC1	!	1	A	Q	a	q	SP	'	ˇ	±	Á	Ň	á	ň
<u>2</u>	<u>0010</u>			"	2	B	R	b	r	,	'	ˇ	˘	Â	Ň	â	ň
<u>3</u>	<u>0011</u>		DC3	#	3	C	S	c	s	SP	"	ł	ł	Ǻ	Ó	ǻ	ó
<u>4</u>	<u>0100</u>			\$	4	D	T	d	t	„	"	ˆ	'	Ä	Ô	ä	ô
<u>5</u>	<u>0101</u>			%	5	E	U	e	u	…	•	À	μ	Í	Õ	í	õ
<u>6</u>	<u>0110</u>			&	6	F	V	f	v	†	-	ı	¶	Ć	Ö	ć	ö
<u>7</u>	<u>0111</u>			'	7	G	W	g	w	‡	-	Š	•	Ç	×	ç	÷
<u>8</u>	<u>1000</u>			(	8	H	X	h	x	SP	SP	¨	˘	Č	Ř	č	ř
<u>9</u>	<u>1001</u>	HT		)	9	I	Y	i	y	%	™	©	ª	É	Ů	é	ů
<u>A</u>	<u>1010</u>	LF		*	:	J	Z	j	z	Š	š	Š	š	É	Ú	e	ú
<u>B</u>	<u>1011</u>		ESC	+	;	K	[	k	{	<	>	«	»	Ě	Ů	ě	ů
<u>C</u>	<u>1100</u>	<u>FF</u>	<u>FS</u>	,	<	L	\			Š	š	¬	ł	Ě	Ü	ě	ü
<u>D</u>	<u>1101</u>	<u>CR</u>	<u>GS</u>	-	=	M	]	m	}	ř	ř	-	˘	Í	Ý	í	ý
<u>E</u>	<u>1110</u>		RS	.	>	N	^	n	~	Ž	ž	®	ı	Î	Ť	î	ť
<u>F</u>	<u>1111</u>			/	?	O	_	o	SP	Ž	ž	Ž	ž	Ď	ß	d'	'

[Notes]

- [SP] indicates space.
- [CR] will be disregarded.
- Printer operation cannot be guaranteed if a blank code in the controlling codes having the equal to or smaller than " [1F] h" number (indicated by the combination of two hexadecimal numbers) or an undefined code is transmitted to the printer.
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- This code table indicates simplified symbols and does not represent the actual printing result by the printer.

3.6 CODE PAGE 1251

	HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
HEX	BIN	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	NUL		SP	0	@	P	`	p	h	h	SP	°	А	Р	а	р
1	0001		DC1	!	1	А	Q	a	q	í	'	ý	±	Б	С	б	с
2	0010			"	2	В	Р	b	r	,	'	ý	І	В	Т	в	т
3	0011		DC3	#	3	С	Ѕ	c	s	í	"	Ј	і	Г	У	г	у
4	0100			\$	4	Д	Т	d	t	,	"	ѡ	г	Д	Ф	д	ф
5	0101			%	5	Е	U	e	u	...	•	Г	μ	Е	Х	е	х
6	0110			&	6	Ғ	V	f	v	†	-	!	¶	Ж	Ц	ж	ц
7	0111			'	7	G	W	g	w	†	-	§	•	З	Ч	з	ч
8	1000			(	8	H	X	h	x	€	SP	Ё	ё	И	Ш	и	ш
9	1001	HT		)	9	I	Y	i	y	%	™	©	№	Й	Щ	й	щ
A	1010	LF		*	:	J	Z	j	z	Љ	љ	Е	е	К	Ъ	к	ъ
B	1011		ESC	+	;	K	[	k	{	<	>	«	»	Л	Ы	л	ы
C	1100	FF	FS	,	<	L	\			Њ	њ	¬	ј	М	Ь	м	ь
D	1101	CR	GS	-	=	M	]	m	}	Ќ	ќ	-	Š	Н	Э	н	э
E	1110		RS	.	>	N	^	n	~	ћ	ћ	®	š	О	Ю	о	ю
F	1111			/	?	О	_	о	SP	Ў	ў	Ї	ї	П	Я	п	я

[Notes]

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3.7 CODE PAGE 1252

	HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
HEX	BIN	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	NUL		SP	0	@	P	`	p	€	SP	SP	°	À	Ð	à	ð
1	0001		DC1	!	1	A	Q	a	q	SP	'	i	±	Á	Ñ	á	ñ
2	0010			"	2	B	R	b	r	,	'	ø	²	Â	Ò	â	ò
3	0011		DC3	#	3	C	S	c	s	f	"	£	³	Ã	Ó	ã	ó
4	0100			\$	4	D	T	d	t	,,	"	¤	´	Ä	Ô	ä	ô
5	0101			%	5	E	U	e	u	---	•	¥	µ	Å	Õ	å	õ
6	0110			&	6	F	V	f	v	†	-	!	¶	Æ	Ö	æ	ö
7	0111			'	7	G	W	g	w	‡	-	§	•	Ç	×	ç	÷
8	1000			(	8	H	X	h	x	^	~	¨	,	È	Ø	è	ø
9	1001	HT		)	9	I	Y	i	y	%	™	©	'	É	Ù	é	ù
A	1010	LF		*	:	J	Z	j	z	Š	š	à	ó	Ê	Ú	ê	ú
B	1011		ESC	+	;	K	[	k	{	<	>	«	»	Ë	Û	ë	û
C	1100	FF	FS	,	<	L	\			Œ	œ	¬	¼	Ì	Ü	ì	ü
D	1101	CR	GS	-	=	M	]	m	}	SP	SP	-	½	Í	Ý	í	ý
E	1110		RS	.	>	N	^	n	~	Ž	ž	®	¾	Î	Þ	î	þ
F	1111			/	?	O	_	o	SP	SP	ÿ	¯	¿	Ï	ß	ï	ÿ

[Notes]

- [SP] indicates space.
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3.8 CODE PAGE 1254

	HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
HEX	BIN	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	NUL		SP	0	@	P	`	p	€	SP	SP	°	À	Ğ	à	ğ
<u>1</u>	<u>0001</u>		DC1	!	1	A	Q	a	q	SP	'	i	±	Á	Ñ	á	ñ
<u>2</u>	<u>0010</u>			"	2	B	R	b	r	,	'	ø	²	Â	Ò	â	ò
<u>3</u>	<u>0011</u>		DC3	#	3	C	S	c	s	f	"	£	³	Ã	Ó	ã	ó
<u>4</u>	<u>0100</u>			\$	4	D	T	d	t	,	"	¤	'	Ä	Ô	ä	ô
<u>5</u>	<u>0101</u>			%	5	E	U	e	u	...	•	¥	μ	Å	Õ	å	õ
<u>6</u>	<u>0110</u>			&	6	F	V	f	v	†	-	!	¶	Æ	Ö	æ	ö
<u>7</u>	<u>0111</u>			'	7	G	W	g	w	‡	-	§	•	Ç	×	ç	÷
<u>8</u>	<u>1000</u>			(	8	H	X	h	x	^	~	"	,	È	Ø	è	ø
<u>9</u>	<u>1001</u>	HT		)	9	I	Y	i	y	%	™	©	'	É	Ù	é	ù
<u>A</u>	<u>1010</u>	LF		*	:	J	Z	j	z	Š	š	à	ó	Ê	Ú	ê	ú
<u>B</u>	<u>1011</u>		ESC	+	;	K	[	k	{	<	>	«	»	Ë	Û	ë	û
<u>C</u>	<u>1100</u>	<u>FF</u>	<u>FS</u>	,	<	L	\			Œ	œ	¬	¼	Ì	Ü	ì	ü
<u>D</u>	<u>1101</u>	<u>CR</u>	<u>GS</u>	-	=	M	]	m	}	SP	SP	-	½	Í	İ	í	ı
<u>E</u>	<u>1110</u>		RS	.	>	N	^	n	~	SP	SP	®	¾	Î	Ş	î	ş
<u>F</u>	<u>1111</u>			/	?	O	_	o	SP	SP	ÿ	-	¿	Ï	ß	ï	ÿ

[Notes]

- [SP] indicates space.
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- Printer operation cannot be guaranteed if a blank code in the controlling codes having the equal to or smaller than " [1F] h" number (indicated by the combination of two hexadecimal numbers) or an undefined code is transmitted to the printer.
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